AZURE DATABASE ADMINISTRATOR

300 Real-time FAQs with Answers

Scenario-based interview & operations reference

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13 domains • 300 non-overlapping questions • production-grade guidance

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How to Use This Reference

This reference contains 300 real-time, scenario-based questions and point-wise answers spanning the full remit of an Azure Database Administrator across PaaS (Azure SQL Database, Managed Instance) and IaaS (SQL Server on Azure VM).

Questions are grouped into 13 domains. Each answer is written as concise, actionable bullet points reflecting how the situation is handled in production - not textbook definitions.

Use it for interview preparation, on-call runbook thinking, and validating your own operational approach. Where a decision is involved, the answer explains the reasoning, not just the button to press.

> **Reading tip**
> Answers assume a real environment: least-privilege security, HA/DR by design, cost-awareness, and validation before cutover. Adapt thresholds and tiers to your workload.

Domain Summary

#	Domain	Questions
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TOTAL		**300**

1. Azure SQL Database — Deployment, Service Tiers & Provisioning

DOMAIN 1

Azure SQL Database — Deployment, Service Tiers & Provisioning

25 real-time scenarios

Q1. A customer-facing web app needs a new transactional database with unpredictable scaling. Which Azure SQL Database purchasing model do you pick and why?

Answer:

- Choose the vCore model over DTU — it decouples compute and storage and maps directly to on-prem CPU/memory sizing.
- Use General Purpose for balanced, budget-friendly workloads, or Business Critical when local-SSD low latency and read replicas are needed.
- Enable the Serverless compute tier so compute auto-scales and auto-pauses during idle periods to cut cost.
- vCore also unlocks Azure Hybrid Benefit, reusing existing SQL Server licenses for up to ~55% savings.

Q2. A stakeholder asks the difference between DTU and vCore models during a migration proposal. How do you explain it?

Answer:

- DTU bundles compute, memory, and I/O into one blended metric — simple but opaque and hard to right-size.
- vCore exposes vCores, memory, and storage independently for predictable sizing and cost transparency.
- vCore supports Hybrid Benefit and reserved capacity; DTU does not.
- Recommend vCore for all new production; keep DTU only for small, stable legacy databases already on it.

Q3. Your dev/test databases sit idle overnight and on weekends, yet you still pay full compute. What do you configure?

Answer:

- Move them to the Serverless compute tier in the General Purpose vCore model.
- Set an auto-pause delay (e.g., 60 minutes) so compute pauses when idle and you pay only for storage.
- Configure min and max vCores so compute auto-scales within a defined range under load.
- Note the first connection after a pause incurs a warm-up delay — acceptable for dev/test, evaluate carefully for prod.

Q4. A single database periodically needs far more compute during month-end batch runs. How do you handle scaling without downtime?

Answer:

- Scale the vCore or DTU tier up before the batch window using Azure CLI, PowerShell, or an automation runbook.
- Scaling is an online operation — a brief reconnect may occur at the very end of the switch-over.
- Automate the scale-up/scale-down with an Elastic Job or Logic App on a schedule tied to the batch calendar.
- Consider Hyperscale or Serverless if the spikes are frequent and unpredictable rather than scheduled.

Q5. Management wants a rough cost forecast before you provision. What factors do you present?

Answer:

- Service tier and vCore/DTU count drive the base compute cost.
- Provisioned storage, backup storage beyond the included amount, and long-term retention add to it.
- Zone-redundancy, read-scale replicas, and geo-replication each multiply compute cost.
- Reserved capacity (1 or 3 years) and Azure Hybrid Benefit reduce the compute portion significantly.

Q6. A team accidentally provisioned a database in the wrong region. What is your remediation path?

Answer:

- You cannot move a database between regions in place — you must copy or restore into the target region.
- Use database copy or geo-restore to create the database in the correct region.
- Update connection strings and firewall/private-endpoint configuration for the new server.
- Decommission the mis-placed database only after validating the new one, then update IaC templates to prevent recurrence.

Q7. You must enforce that all new SQL databases in the subscription are zone-redundant. How do you guarantee it?

Answer:

- Author an Azure Policy that audits or denies non-zone-redundant SQL databases.
- Assign the policy at the subscription or management-group scope.
- Bake zone-redundancy into your Bicep/ARM/Terraform modules so IaC deployments comply by default.
- Zone redundancy is available in Business Critical, Premium, and Hyperscale tiers in supported regions.

Q8. A developer needs a copy of production for testing but must not touch live data. What is the safest approach?

Answer:

- Use the database copy operation to create an independent, transactionally consistent copy.
- Or restore a point-in-time copy to a new database name so production is untouched.
- Mask or scrub sensitive columns in the copy before handing it to developers to meet compliance.
- Place the copy on a lower service tier to control cost, since test load is lighter.

Q9. Your organization standardizes deployments with infrastructure-as-code. Which tools do you use for Azure SQL?

Answer:

- Use Bicep or ARM templates for native Azure resource definitions, or Terraform for multi-cloud estates.
- Manage schema/state with DACPAC/BACPAC via SqlPackage in the CI/CD pipeline.
- Store templates in Git and deploy through Azure DevOps or GitHub Actions pipelines.
- Parameterize tier, region, and firewall rules so the same module serves dev, test, and prod.

Q10. What is the practical difference between a logical server and the databases under it in Azure SQL Database?

Answer:

- The logical server is a management and connectivity container — it has no compute of its own.
- It holds server-level firewall rules, Azure AD admin, auditing settings, and failover groups.
- Each database has its own service tier, compute, and storage independent of siblings.
- Connection strings target the server FQDN, then resolve to the specific database.

Q11. You need to consolidate 40 small SaaS-tenant databases with varying, bursty load. What design do you recommend?

Answer:

- Place them in an Elastic Pool so they share a common pool of eDTUs/vCores.
- Pooling smooths cost because tenants rarely peak simultaneously.
- Set per-database min/max caps so no single tenant starves the others.
- Monitor pool utilization and split into multiple pools if aggregate demand outgrows one.

Q12. A compliance auditor asks where your Azure SQL databases physically reside. How do you answer confidently?

Answer:

- Data resides in the Azure region chosen at provisioning; confirm via the resource's location property.
- Geo-replicated or geo-backup copies may reside in the paired region — document that in the data-residency record.
- Use Azure Policy to restrict allowed regions and prevent provisioning outside approved geographies.
- For strict residency, disable geo-redundant backup storage and use zone- or locally-redundant storage instead.

Q13. Your app throws 'database is at capacity' style throttling errors during peaks. What is your triage order?

Answer:

- Check whether you have hit the DTU/vCore ceiling or the max concurrent worker/session limit for the tier.
- Review Query Store and resource metrics to confirm CPU, log I/O, or worker exhaustion.
- Scale up the tier or move to Serverless/Hyperscale if the ceiling is genuine.
- Address inefficient queries first — throttling is often a symptom of missing indexes or runaway plans.

Q14. How do you decide between Azure SQL Database and Azure SQL Managed Instance for a lift-and-shift?

Answer:

- Choose Managed Instance when the app needs instance-scoped features: SQL Agent, cross-database queries, CLR, Service Broker, or linked servers.
- Choose Azure SQL Database for single-database or elastic-pool workloads that don't need instance features.
- Managed Instance offers near-100% surface-area compatibility with on-prem SQL Server.
- Azure SQL Database gives finer-grained scaling and Serverless/Hyperscale options.

Q15. A single database has grown past 4 TB and continues expanding rapidly. What tier addresses this?

Answer:

- Move it to the Hyperscale service tier, which supports up to 128 TB and beyond.
- Hyperscale decouples compute, log, and page servers, enabling near-instant scaling and fast backups.
- Backups are snapshot-based and virtually instantaneous regardless of size.
- You can add read-scale replicas for reporting without impacting the primary.

Q16. You must apply a consistent maintenance window to avoid updates during business hours. How?

Answer:

- Configure a Maintenance Window on the database/pool to move planned maintenance to off-peak times.
- Choose a system-defined or custom window (e.g., weekday nights or weekends).
- Maintenance windows reduce the impact of planned reconfigurations and patching.
- Combine with retry logic in the app to absorb the brief reconfiguration events gracefully.

Q17. Connections intermittently drop with transient errors. What is the correct application-side pattern?

Answer:

- Implement retry logic with exponential backoff for transient error numbers (e.g., 40197, 40501, 40613, 49918).
- Use a resilient data-access library (EF Core execution strategy, Polly, or the SQL client retry provider).
- Keep connection pools healthy and open connections late, close early.
- Transient errors are expected in PaaS during reconfiguration — the app, not the DBA, must tolerate them.

Q18. You need read-only reporting offloaded from the primary without extra licensing. What do you enable?

Answer:

- In Business Critical/Premium, enable read scale-out to route read-only connections to a free replica.
- Set ApplicationIntent=ReadOnly in the connection string to target the replica.

- This offloads reporting/analytics from the read-write primary at no extra compute cost.
- For heavier reporting, add named read replicas (Hyperscale) instead.

Q19. A new database must be deployed identically across three regions for a global app. Approach?

Answer:

- Parameterize a single IaC module and deploy to each region with region-specific values.
- Use a failover group or geo-replication to keep secondaries in sync where needed.
- Ensure consistent firewall/private-endpoint and Azure AD admin configuration per region.
- Front the regional endpoints with Traffic Manager or Front Door for latency-based routing.

Q20. Finance asks why last month's Azure SQL bill spiked. Where do you look first?

Answer:

- Open Cost Management and filter by the SQL resource, then break down by meter (compute, storage, backup, geo-replication).
- Check for tier scale-ups, added replicas, or long-term retention backups accumulating.
- Look for Serverless databases that stopped auto-pausing due to constant connections.
- Recommend reserved capacity or Hybrid Benefit if the workload is steady.

Q21. You inherit an environment with no tagging. Why does it matter for a DBA and what do you do?

Answer:

- Tags drive cost allocation, ownership, environment identification, and automation targeting.
- Apply a tagging standard (owner, environment, cost-center, app) via Azure Policy with enforcement.
- Backfill tags on existing resources with a PowerShell/CLI script.
- Use tags to scope automation runbooks (e.g., pause all dev databases nightly).

Q22. A database must never be accidentally deleted. What controls do you apply?

Answer:

- Apply a CanNotDelete resource lock at the database or resource-group scope.
- Restrict delete permissions via RBAC to a small set of admins.
- Rely on point-in-time restore as a recovery net if a deletion still occurs.
- Deleted servers/databases can be recovered within the retention window via geo-restore of backups.

Q23. The business wants sub-second failover for a critical single database. Which tier and feature?

Answer:

- Use Business Critical with zone redundancy for automatic in-region HA across availability zones.
- Zone-redundant HA keeps synchronized replicas in separate zones with automatic failover.
- Add a failover group for cross-region DR on top of in-region HA.
- Test failover regularly and confirm the app tolerates the reconnect.

Q24. How do you right-size a newly migrated database whose on-prem metrics you already have?

Answer:

- Map peak CPU cores and memory to an equivalent vCore tier, adding headroom.
- Use the Data Migration Assistant/SKU recommendation tooling to suggest a tier from a workload trace.
- Start slightly conservative, then monitor Query Store and resource metrics for a week and adjust.
- Scaling is online, so it's cheap to correct after observing real load.

Q25. A team asks whether they can run cross-database joins in Azure SQL Database. What's your guidance?

Answer:

- Native cross-database queries aren't supported the same way as on-prem in Azure SQL Database.
- Use Elastic Query (external tables) for cross-database read scenarios, or consolidate into one database.

- If heavy cross-database logic is core to the app, recommend Managed Instance instead.
- Re-architect toward a single database or a data-integration layer where feasible.

2. Azure SQL Managed Instance — Administration & Operations

DOMAIN 2

Azure SQL Managed Instance — Administration & Operations

25 real-time scenarios

Q26. A legacy app uses SQL Agent jobs, linked servers, and cross-database queries. Which Azure target fits and why?

Answer:

- Azure SQL Managed Instance — it supports SQL Agent, linked servers, cross-database queries, CLR, and Service Broker.
- It offers near-full surface-area compatibility with on-prem SQL Server, minimizing code changes.
- You get instance-scoped administration similar to a boxed SQL Server.
- Azure SQL Database would force re-architecting those instance-level features.

Q27. You must deploy a Managed Instance but security requires it be isolated from the internet. How do you architect networking?

Answer:

- Deploy the MI into a dedicated delegated subnet within your VNet.
- The subnet must be delegated to Managed Instance and sized with enough addresses for scaling.
- Apply an NSG and route table per the MI networking requirements; avoid blocking management traffic.
- Use a private endpoint or VNet peering/VPN for app connectivity, keeping it off the public internet.

Q28. Provisioning a Managed Instance is taking hours. Is that normal, and what do you tell stakeholders?

Answer:

- Yes — first MI deployment into a new subnet can take a few hours because a virtual cluster is created.
- Subsequent instances in the same subnet deploy faster.
- Scaling operations and some maintenance also involve the virtual cluster and can take time.
- Plan provisioning and scaling windows accordingly and communicate expected durations up front.

Q29. You need to migrate 20 databases from on-prem to Managed Instance with minimal downtime. Best method?

Answer:

- Use the Log Replay Service (LRS) or Azure Database Migration Service in online mode.
- Restore full/differential/log backups from Azure Blob Storage and continuously replay logs until cutover.
- Perform a final log restore and cutover during a short maintenance window.
- Native RESTORE from URL works because MI supports restoring standard SQL Server backups.

Q30. A DBA wants to move a database between two Managed Instances. What's the cleanest method?

Answer:

- Take a native COPY_ONLY backup to Blob Storage and RESTORE onto the target MI.
- Or use cross-instance point-in-time restore if both are in the same region/subscription.
- For continuous sync, use the Log Replay Service.
- Managed Instance supports native backup/restore, unlike Azure SQL Database.

Q31. Your Managed Instance is running low on storage. How do you respond and prevent recurrence?

Answer:

- Increase the reserved storage on the instance (online operation within tier limits).
- Identify large databases/log growth and reclaim space; check for unmanaged log growth.

- Set up storage alerts in Azure Monitor to warn before hitting the ceiling.
- General Purpose scales storage per-file; Business Critical has a fixed local-SSD envelope to plan around.

Q32. The business asks for cross-region DR for a Managed Instance. What do you configure?

Answer:

- Set up an auto-failover group pairing the primary MI with a secondary MI in the paired region.
- The failover group replicates all databases and provides a read-write and read-only listener endpoint.
- Configure the app to use the failover-group listener so failover is transparent.
- Test planned and forced failover and document RTO/RPO.

Q33. You need SQL Agent jobs to run maintenance on the Managed Instance. Any caveats vs on-prem?

Answer:

- SQL Agent is supported on MI, so Ola Hallengren maintenance solution works well.
- Some subsystems differ; database mail and certain agent features have MI-specific behavior — validate them.
- Schedule index/stats maintenance and integrity checks like on-prem.
- Keep job history and alerting wired into Azure Monitor for visibility.

Q34. Business Critical vs General Purpose on Managed Instance — how do you choose?

Answer:

- General Purpose: remote storage, single compute node, cost-effective, good for most OLTP.
- Business Critical: local SSD, built-in Always On replicas, low latency, and a free readable secondary.
- Choose Business Critical for latency-sensitive, high-throughput, or read-scale needs.
- Both support zone redundancy in supported regions for higher availability.

Q35. How do you connect on-prem apps to a Managed Instance during a hybrid phase?

Answer:

- Use ExpressRoute or a site-to-site VPN to bridge on-prem to the MI's VNet.
- Enable a public endpoint only if strictly required, tightly firewalled — private connectivity is preferred.
- Ensure DNS resolves the MI FQDN to the private IP.
- Validate the required ports and MI networking rules are open end to end.

Q36. A restore of a large database onto MI is slow. What influences restore time and how do you optimize?

Answer:

- Restore speed depends on backup size, compression, storage tier, and instance compute.
- Use compressed backups and place them in the same region's Blob Storage.
- General Purpose remote storage can be slower than Business Critical local SSD for large restores.
- Split very large databases and parallelize where feasible; scale up compute temporarily.

Q37. You must audit all logins and DDL on a Managed Instance for compliance. How?

Answer:

- Enable SQL Managed Instance auditing writing to a storage account, Log Analytics, or Event Hub.
- Capture successful/failed logins and schema changes via audit action groups.
- Route audit logs to Log Analytics for KQL querying and alerting.
- Set retention on the storage/workspace to satisfy the compliance mandate.

Q38. A tempdb-heavy workload struggles on Managed Instance. What are your levers?

Answer:

- MI configures multiple tempdb files by default; verify sizing and file count fit the workload.
- Move to Business Critical for local-SSD tempdb performance if latency-bound.
- Reduce tempdb pressure by fixing spills, sorts, and excessive temp-table usage in queries.

- Scale compute up if tempdb contention correlates with CPU/memory pressure.

Q39. You need to patch or the platform patches MI — how do you minimize business impact?

Answer:

- Configure a Maintenance Window to schedule planned updates into off-peak hours.
- Ensure apps use the failover-group listener and retry logic to survive brief failovers during patching.
- Monitor the health after maintenance events via Azure Monitor.
- Communicate the window to stakeholders; MI patching is largely automated by the platform.

Q40. Cross-database transactions are failing on MI in some cases. What do you check?

Answer:

- MI supports cross-database queries, but distributed transactions require Server Trust Group configuration.
- Create a Server Trust Group to enable distributed transactions across MIs.
- For MSDTC-style needs, confirm the specific pattern is supported and configured.
- Validate linked-server and security context settings for cross-instance calls.

Q41. How do you manage instance-level logins and permissions consistently across MIs?

Answer:

- Prefer Azure AD (Microsoft Entra) authentication and group-based access for centralized control.
- Script server-level logins/roles so they are reproducible across primary and DR instances.
- In failover groups, ensure logins exist on both instances to avoid orphaned users after failover.
- Use contained-style patterns where possible and document the RBAC/SQL permission model.

Q42. A team wants to know if they can use Database Mail on MI for job alerts. What's the answer?

Answer:

- Database Mail is supported on Managed Instance with some configuration specifics.
- Configure an SMTP relay reachable from the MI subnet (e.g., an authenticated relay).
- Wire SQL Agent job failure notifications to Database Mail operators.
- Also forward alerts to Azure Monitor action groups for redundancy.

Q43. You must reduce MI cost for a steady 24/7 workload. Options?

Answer:

- Apply Azure Hybrid Benefit to reuse existing SQL Server core licenses.
- Purchase reserved capacity (1 or 3 years) for the compute.
- Right-size vCores based on observed utilization rather than peak-of-peak.
- Consolidate low-utilization databases onto fewer instances where isolation allows.

Q44. A restore must land at a specific point in time after a bad deployment on MI. How?

Answer:

- Use point-in-time restore to a new database name within the retention window.
- Restore to just before the bad deployment, validate, then swap or repoint the app.
- MI PITR uses automated backups; confirm your retention covers the needed timestamp.
- Keep the original database until validation completes to preserve rollback options.

Q45. Users report slow connections to MI from a specific office. What network angles do you investigate?

Answer:

- Check the network path (VPN/ExpressRoute) latency and packet loss from that site.
- Verify DNS resolves to the private IP and there's no hairpin through the internet.
- Confirm proxy vs redirect connection type; redirect lowers latency but needs specific ports open.
- Inspect NSG/firewall rules along the route for that subnet.

Q46. What's your approach to capacity planning for a Managed Instance over the next year?

Answer:

- Trend vCore CPU, memory, storage, and IOPS from Azure Monitor over months.
- Project growth from business forecasts (new tenants, data retention policy changes).
- Plan tier/vCore and storage increases ahead of thresholds, accounting for scaling durations.
- Reserve capacity once the steady-state size is known to lock in savings.

Q47. How do you validate a Managed Instance migration was successful before cutover?

Answer:

- Compare row counts, schema, and object inventories between source and target.
- Run application smoke tests and key business queries against the MI.
- Validate logins, jobs, linked servers, and permissions carried over.
- Check performance against baselines and confirm backups and DR are configured.

Q48. A Managed Instance shows unexpected high CPU after migration. First moves?

Answer:

- Enable/inspect Query Store to find regressed or high-cost plans versus the on-prem baseline.
- Check compatibility level and cardinality-estimator changes that alter plans post-migration.
- Update statistics and confirm indexes migrated correctly.
- Use automatic tuning / plan forcing to stabilize regressed queries.

Q49. You need to give developers read-only access to production MI for troubleshooting. How, safely?

Answer:

- Grant a read-only role via Azure AD group membership mapped to a SQL role.
- Scope permissions to specific databases and deny data-modification and DDL.
- Consider a readable secondary (Business Critical) so troubleshooting doesn't load the primary.
- Audit their access and time-box it via just-in-time access where possible.

Q50. Explain how backups work on Managed Instance and what you as DBA must still do.

Answer:

- Automated full, differential, and log backups run by the platform enabling PITR.
- You configure PITR retention (up to 35 days) and optional long-term retention for years.
- You can also take COPY_ONLY native backups to Blob for portability without breaking the chain.
- Regularly test restores — automated backups are only useful if restores are validated.

3. SQL Server on Azure Virtual Machines (IaaS)

DOMAIN 3

SQL Server on Azure Virtual Machines (IaaS)

20 real-time scenarios

Q51. An app requires OS-level access and features PaaS can't offer. You choose SQL on Azure VM — what VM sizing rules apply?

Answer:

- Pick memory-optimized VM families (e.g., Edsv5/Ebdsv5) for SQL to get high memory-to-vCPU ratios.
- Ensure the VM series supports enough data disks and IOPS/throughput for your I/O profile.
- Size cores to your SQL licensing to avoid paying for unused compute (or use AHB).
- Enable accelerated networking and constrained-core VMs to optimize licensing where supported.

Q52. You must lay out storage for a high-I/O SQL Server VM. What's your disk design?

Answer:

- Separate data, log, and tempdb onto different Premium SSD/Ultra Disk volumes.
- Stripe multiple disks with Storage Spaces to aggregate IOPS/throughput as needed.
- Put tempdb on the local ephemeral SSD when the VM series provides one, for lowest latency.
- Format data/log disks with 64 KB allocation unit size and enable read caching on data, none on log.

Q53. The SQL Server IaaS Agent extension is mentioned in a review. What value does it add?

Answer:

- It enables automated patching, automated backup, and Azure Key Vault integration for the VM.
- It surfaces SQL best-practice assessments and license-type management (PAYG vs AHB) in the portal.
- It allows storage configuration and monitoring integration.
- Register existing SQL VMs with the extension to gain these management features.

Q54. You need HA for SQL on Azure VMs equivalent to on-prem Always On AGs. How do you build it?

Answer:

- Deploy 2+ SQL VMs across Availability Zones and configure an Always On Availability Group.
- Use an Azure Load Balancer (or distributed network name) for the AG listener.
- Place VMs in a proximity/zone layout that balances latency and fault isolation.
- Configure the Windows Failover Cluster with a cloud witness in Azure Storage for quorum.

Q55. Quorum for a two-node WSFC in Azure — what witness do you use and why?

Answer:

- Use a Cloud Witness backed by an Azure Storage account.
- It's cheap, zone-resilient, and avoids maintaining a third VM as a file-share witness.
- It provides the tie-breaking vote for a two-node cluster to prevent split-brain.
- Secure the storage account and rotate keys per policy.

Q56. Backups for SQL on Azure VM — what are your options and trade-offs?

Answer:

- Use Azure Backup for SQL Server in Azure VM for managed, policy-driven backups with central restore.
- Or run native SQL backups to URL (Blob Storage) for full control and portability.
- Automated Backup via the IaaS extension is simplest but less flexible.
- Whichever you choose, test restores and monitor backup success/failure via alerts.

Q57. You want to reduce SQL licensing cost on Azure VMs. Levers?

Answer:

- Apply Azure Hybrid Benefit to use existing SQL Server licenses with Software Assurance.
- Use constrained-core VM sizes to keep high memory while licensing fewer cores.
- Right-size the edition — Standard vs Enterprise — to actual feature needs.
- Consider reserved VM instances for steady workloads to cut compute cost.

Q58. Patching Windows and SQL on many SQL VMs is error-prone. How do you standardize it?

Answer:

- Use Automated Patching via the SQL IaaS extension with a defined maintenance window.
- Or orchestrate with Azure Update Manager across the fleet.
- For AGs, patch secondaries first, fail over, then patch the former primary to avoid downtime.
- Validate app health after each node and keep a rollback plan.

Q59. TempDB performance is poor on a SQL VM. What do you tune?

Answer:

- Move tempdb to the local ephemeral NVMe/SSD disk the VM provides.
- Configure multiple equally-sized tempdb data files to reduce allocation contention.
- Pre-size tempdb to avoid autogrowth during peaks.
- The IaaS extension can auto-place tempdb on local SSD on supported series.

Q60. A SQL VM shows high I/O latency. Where do you look?

Answer:

- Check disk caching settings (read caching on data, none on log) and disk type (Premium/Ultra).
- Confirm you haven't hit VM-level or disk-level IOPS/throughput caps.
- Stripe additional disks or move to Ultra Disk for higher, tunable throughput.
- Correlate with SQL wait stats (PAGEIOLATCH) to confirm storage is the bottleneck.

Q61. You must secure a SQL VM's management and data plane. What's your checklist?

Answer:

- No public RDP/SQL ports — use Bastion/Just-in-Time VM access for management.
- Restrict SQL port exposure with NSGs; keep it on a private subnet.
- Enable TDE and store keys in Azure Key Vault via the extension.
- Enable Microsoft Defender for SQL on machines for vulnerability assessment and threat alerts.

Q62. How do you provide DR across regions for SQL on Azure VMs?

Answer:

- Extend the Always On AG with an asynchronous replica in the paired region.
- Or use Azure Site Recovery for VM-level replication if AG isn't feasible.
- Automate DNS/listener updates and test failover regularly.
- Document RTO/RPO and confirm networking (VNet peering/DNS) works cross-region.

Q63. Autogrowth events are causing stalls on a SQL VM. Remediation?

Answer:

- Pre-size data and log files to expected sizes to avoid frequent autogrowth.
- Set sensible fixed-size growth increments rather than percentage growth.
- Enable instant file initialization (grant Perform Volume Maintenance Tasks) for fast data-file growth.
- Monitor free space and alert before files fill.

Q64. You need to migrate an on-prem SQL Server to Azure VM with minimal downtime. Method?

Answer:

- Set up an Always On AG or log shipping from on-prem to the Azure VM, then fail over at cutover.
- Or use Azure Migrate for VM-level lift-and-shift when downtime tolerance is higher.
- Restore-from-URL of backups is simplest for smaller databases.
- Validate performance, connectivity, and DR post-migration.

Q65. Memory settings on a fresh SQL VM are causing OS pressure. What do you configure?

Answer:

- Set SQL Server max server memory to leave headroom for the OS and other processes.
- Enable Lock Pages in Memory for the SQL service account on supported editions.
- Confirm the VM has a memory-optimized ratio suitable for the workload.
- Monitor available MB and page life expectancy to validate the setting.

Q66. How do you monitor a fleet of SQL VMs centrally?

Answer:

- Onboard VMs to Azure Monitor with the agent and enable VM insights.
- Collect SQL metrics/logs into Log Analytics for KQL queries and alerts.
- Use the SQL best-practices assessment from the IaaS extension for drift detection.
- Build dashboards and alert rules for CPU, memory, disk latency, and AG health.

Q67. A stakeholder asks why not just always use PaaS. When is SQL on Azure VM the right call?

Answer:

- When you need OS access, unsupported PaaS features, specific SQL versions/editions, or third-party agents.
- When you require full control over patching cadence and configuration.
- For lift-and-shift where re-architecting for PaaS isn't justified.
- Trade-off: you own OS/SQL patching, HA/DR setup, and backups that PaaS would manage.

Q68. Your AG listener isn't reachable from clients on Azure VMs. What's the usual cause?

Answer:

- The Azure Load Balancer for the listener isn't configured with the right floating-IP/health-probe.
- Verify the probe port, backend pool, and the listener IP match the AG configuration.
- Confirm NSGs allow the listener and probe ports.
- Alternatively use a Distributed Network Name (DNN) listener to avoid the load balancer entirely.

Q69. You must ensure consistent SQL configuration across new SQL VMs. Approach?

Answer:

- Build from a hardened, marketplace SQL image or a custom golden image.
- Apply configuration with DSC/Ansible or the IaaS extension for repeatable settings.
- Codify VM + disk + SQL settings in IaC (Bicep/Terraform).
- Run the best-practices assessment to catch configuration drift.

Q70. How do you handle Windows/SQL time-zone and collation gotchas on Azure VMs during migration?

Answer:

- Match the target instance collation to the source to avoid comparison/sort issues.
- Set the VM time zone consistently and be explicit about UTC in the app where possible.
- Validate SQL Agent job schedules against the VM's local time.
- Test date-sensitive queries and jobs after migration.

4. High Availability & Disaster Recovery

DOMAIN 4

High Availability & Disaster Recovery

30 real-time scenarios

Q71. The business defines RTO of 1 hour and RPO of 5 minutes for a critical Azure SQL Database. What do you design?

Answer:

- Use a failover group (or active geo-replication) to a paired region for cross-region DR.
- Geo-replication is asynchronous with typically low RPO — validate it meets the 5-minute target under load.
- Automatic failover policy plus a listener endpoint achieves the RTO with minimal manual steps.
- Combine with in-region zone-redundant HA so most failures never require a region failover.

Q72. Explain the difference between active geo-replication and auto-failover groups.

Answer:

- Active geo-replication creates readable secondaries you fail over manually per database.
- Auto-failover groups add automatic failover, a stable listener endpoint, and group-level failover of many databases.
- Failover groups keep the same connection endpoint after failover; geo-replication requires connection-string changes.
- Use failover groups for production DR; use geo-replication for read-scale or granular control.

Q73. After a regional outage, your failover group promoted the secondary. What must you verify post-failover?

Answer:

- Confirm the app reconnected via the listener and is serving traffic from the new primary.
- Check that server-level logins, firewall rules, and auditing exist on the promoted server.
- Validate replication is re-established once the original region recovers.
- Communicate status, then plan a controlled failback during a maintenance window.

Q74. A single database needs protection against an availability-zone failure but not a region failure. Cheapest robust option?

Answer:

- Enable zone-redundant configuration on Business Critical/Premium (or Hyperscale) so replicas span zones.
- Automatic in-region failover across zones handles a zone outage transparently.
- No cross-region cost is incurred, keeping it economical.
- Ensure the app has retry logic to absorb the brief failover reconnect.

Q75. For SQL Server on Azure VMs, walk through building an Always On AG for HA.

Answer:

- Provision SQL VMs across availability zones and join them to a Windows Failover Cluster with a cloud witness.
- Create the AG with synchronous-commit replicas for automatic failover in-region.
- Configure the listener via Azure Load Balancer (or DNN) so clients reconnect automatically.
- Add an async replica in the paired region for DR.

Q76. Synchronous vs asynchronous commit in Always On — how do you decide per replica?

Answer:

- Synchronous commit = zero data loss but adds commit latency; use for in-region HA replicas.
- Asynchronous commit = potential data loss but no latency penalty; use for distant DR replicas.

- Never place a synchronous replica across high-latency links — it throttles the primary.
- Automatic failover requires synchronous commit and the healthy quorum condition.

Q77. Your AG shows a replica in 'Not Synchronizing / Resolving' state. Triage steps?

Answer:

- Check network connectivity and the endpoint between replicas.
- Verify the WSFC is healthy and quorum is intact (witness reachable).
- Inspect the AG dashboard and error log for the specific failure reason.
- Resume data movement on the affected database once the underlying issue is fixed.

Q78. How do you test DR without impacting production for an Azure SQL failover group?

Answer:

- Use a planned failover (friendly failover) during a maintenance window to validate no data loss.
- Time the failover to measure actual RTO and confirm app reconnection.
- Fail back afterward and confirm replication resynchronizes.
- Document results and update the DR runbook with observed metrics.

Q79. A stakeholder confuses HA and DR. How do you clearly distinguish them?

Answer:

- HA protects against local/component failures automatically, usually in-region (zones, replicas).
- DR protects against regional disasters, typically cross-region with defined RTO/RPO.
- HA aims for near-zero downtime; DR accepts some recovery time for a larger-scope event.
- A mature design layers both: zone-redundant HA plus cross-region failover group.

Q80. Read-only reporting is hammering your primary. How do HA replicas help?

Answer:

- Enable read scale-out (Business Critical/Premium) to route ApplicationIntent=ReadOnly to a free replica.
- For geo-replicated secondaries, direct reporting to the readable secondary endpoint.
- On AGs, configure read-only routing to send read-intent connections to secondaries.
- This offloads the primary while keeping HA/DR intact.

Q81. What quorum model do you use for a 3-node AG spread across two Azure regions?

Answer:

- Use Node and Cloud Witness (or dynamic quorum) to survive site loss without split-brain.
- Assign votes carefully so the primary site retains quorum when the DR site is unreachable.
- Use a cloud witness in a third region/zone to break ties.
- Test quorum behavior by simulating node and site failures.

Q82. Failback after a DR event is risky. How do you plan it?

Answer:

- Ensure the original primary is fully resynchronized before failing back.
- Schedule failback in a maintenance window using a planned (no-data-loss) failover.
- Validate app connectivity, replication direction, and monitoring after failback.
- Update the runbook and confirm all logins/permissions are consistent on both sides.

Q83. Your RPO is being missed on an async DR replica. What increases replication lag and how do you fix it?

Answer:

- High primary write volume, network latency/bandwidth limits, or under-sized secondary compute.
- Reduce redo/queue backlog by scaling the secondary compute up.
- Check network throughput between regions; consider ExpressRoute for VMs.

- Monitor the replication lag metric and alert when it exceeds the RPO threshold.

Q84. How do you ensure logins and jobs survive a failover on Azure SQL Managed Instance?

Answer:

- In failover groups, create server-level logins and agent jobs on both instances.
- Use Azure AD groups to avoid orphaned SQL logins after failover.
- Script instance-level objects so both primary and secondary stay in sync.
- Test a failover and confirm jobs and access still function.

Q85. A region is degraded but not fully down; you must decide whether to fail over. What informs the call?

Answer:

- Assess impact: are app errors/timeouts breaching SLA versus the risk of a failover event?
- Check RPO exposure — a forced failover on async replication risks data loss.
- Weigh RTO of failover against expected recovery time of the degraded region.
- Follow the documented decision tree and get the agreed business sign-off.

Q86. Explain geo-restore and when it's your DR tool of choice.

Answer:

- Geo-restore recreates a database from geo-redundant backups into any region.
- Use it when you lack a running geo-secondary but need to recover after a regional loss.
- RPO/RTO are larger than failover groups because it restores from backups.
- It's the safety net for databases not covered by geo-replication.

Q87. How does zone-redundant Hyperscale provide HA differently from Business Critical?

Answer:

- Hyperscale separates compute, log service, and page servers; HA comes from replicating these tiers across zones.
- It provides fast failover and near-instant backups regardless of size.
- Add HA secondary replicas across zones for automatic failover.
- Business Critical uses local-SSD Always On replicas; Hyperscale scales storage far larger.

Q88. A DR runbook hasn't been tested in a year. Why is that a problem and what do you do?

Answer:

- Untested runbooks fail during real incidents due to drift in endpoints, logins, or steps.
- Schedule regular DR drills (e.g., quarterly) with a planned failover.
- Capture RTO/RPO measured during drills and fix gaps.
- Keep the runbook version-controlled and reviewed after every infra change.

Q89. You must minimize connection changes for apps during failover. What's the design principle?

Answer:

- Use a stable listener endpoint — failover-group listener (PaaS) or AG listener (VM).
- The endpoint stays constant while the underlying primary changes.
- Apps only need retry logic, not connection-string rewrites.
- For read traffic, expose the read-only listener/endpoint separately.

Q90. Cross-subscription or cross-tenant DR is requested. What are the considerations?

Answer:

- Failover groups require the paired servers in compatible subscriptions/regions — validate support.
- Ensure RBAC, networking, and Azure AD trust are configured across the boundary.
- For strict isolation, use backup/geo-restore into the target subscription.
- Document the ownership and access model for the DR environment.

Q91. How do you calculate whether synchronous HA is even possible between two sites?

Answer:

- Measure round-trip latency; synchronous commit adds that latency to every transaction commit.
- If RTT is more than a few milliseconds, synchronous commit will hurt throughput.
- Use synchronous only within a metro/zone distance; use async for longer distances.
- Validate with load testing before committing to synchronous replication.

Q92. A failover group failover succeeded but reporting apps broke. Why?

Answer:

- Reporting apps likely pointed at the old read-only secondary endpoint that changed roles.
- Use the failover-group read-only listener so read traffic follows the new topology.
- Ensure ApplicationIntent=ReadOnly connections resolve via the listener.
- Update any hard-coded secondary endpoints to the listener pattern.

Q93. Explain how automatic failover policy decisions are made in Azure SQL failover groups.

Answer:

- The platform monitors health; if the primary is unavailable beyond the grace period, it auto-fails over.
- The read-write grace period balances avoiding false failovers against RTO.
- A longer grace period reduces false positives but increases downtime tolerance.
- Tune the grace period to your RTO and the app's transient-fault tolerance.

Q94. You need HA for a Hyperscale database with heavy read workloads. Design?

Answer:

- Enable zone redundancy for HA and add named read replicas for scale-out reads.
- Route reporting to read replicas via connection intent/endpoints.
- Add a geo-replica for cross-region DR if required.
- Monitor replica lag and scale replicas independently of the primary.

Q95. During a WSFC issue, both AG nodes think they're primary (split-brain risk). How do you prevent this?

Answer:

- Maintain proper quorum with an odd vote count and a witness (cloud witness in Azure).
- Use synchronous commit with automatic failover only under healthy quorum.
- Never manually force quorum without understanding data-loss implications.
- Test node/site isolation scenarios so quorum behaves as designed.

Q96. A team asks whether backups alone satisfy DR. How do you correct that assumption?

Answer:

- Backups enable recovery but with higher RTO/RPO than replication-based DR.
- For strict RTO/RPO, use failover groups/geo-replication or AGs, with backups as the safety net.
- Backups don't provide a hot standby or automatic failover.
- Layer both: geo-redundant backups plus a replication-based failover target.

Q97. How do you validate that your DR target can actually handle production load?

Answer:

- Size the DR replica/instance to match production capacity, not a reduced tier.
- Run a failover drill and drive representative load against the DR primary.
- Monitor CPU, memory, and latency during the drill.
- Fix any under-sizing before relying on the DR site.

Q98. What's your approach to monitoring HA/DR health proactively?

Answer:

- Alert on replication lag, replica health state, and failover events via Azure Monitor.
- Track WSFC/AG health for VM deployments and geo-link health for PaaS.
- Set thresholds tied to your RPO so you're warned before breach.
- Include HA/DR checks in daily operational dashboards.

Q99. A planned Azure maintenance event caused an unexpected failover. Is this a problem?

Answer:

- No — planned maintenance can trigger a brief failover; it's expected in PaaS HA.
- The listener keeps the endpoint stable; the app should retry transparently.
- Configure a maintenance window to control when these occur.
- If the app broke, the fix is retry logic, not disabling HA.

Q100. You must document RTO/RPO commitments for auditors. What do you include?

Answer:

- The chosen HA/DR architecture per database and its measured RTO/RPO from drills.
- Backup retention (PITR and LTR) and geo-redundancy settings.
- Runbook steps, roles/responsibilities, and escalation paths.
- Evidence of the last successful DR test with metrics.

5. Backup, Restore & Recovery

DOMAIN 5

Backup, Restore & Recovery

25 real-time scenarios

Q101. A developer dropped a critical table 20 minutes ago in Azure SQL Database. How do you recover just that data?

Answer:

- Use point-in-time restore to a new database at a timestamp just before the drop.
- Restore to a new name so production stays intact during recovery.
- Extract the table's data from the restored copy and merge it back.
- Confirm the retention window covers the required time before starting.

Q102. Explain the difference between PITR and long-term retention (LTR) in Azure SQL.

Answer:

- PITR uses automated backups to restore to any second within the configured retention (up to 35 days).
- LTR keeps weekly/monthly/yearly full backups for up to 10 years for compliance.
- PITR handles operational recovery; LTR handles regulatory archival.
- Restoring an LTR backup creates a new database from that specific archived full backup.

Q103. Compliance requires 7-year retention of monthly backups. How do you configure it?

Answer:

- Enable Long-Term Retention on the database with a monthly (and possibly yearly) policy set to 7 years.
- LTR backups are stored in geo-redundant storage by default for durability.
- Document the policy and validate a restore from an LTR backup periodically.
- Monitor LTR storage cost as it accumulates over years.

Q104. How do you restore a Managed Instance database to a point in time after a bad batch job?

Answer:

- Use MI point-in-time restore to a new database name within the retention window.
- Choose a timestamp just before the batch job started.
- Validate the restored data, then repoint the app or swap the database.
- Keep the original until validation is complete for rollback safety.

Q105. A restore is needed but the retention window has passed. What options remain?

Answer:

- If LTR was configured, restore from the nearest LTR full backup.
- If neither PITR window nor LTR covers it, the data may be unrecoverable — escalate honestly.
- This is why retention policies must match business/compliance needs proactively.
- Review and extend retention to prevent recurrence.

Q106. You must move a database to another region as a one-off. Which backup-based method?

Answer:

- Use geo-restore (from geo-redundant backups) or database copy into the target region.
- Geo-restore recreates the database in any region from the latest geo-replicated backup.
- Validate and repoint the app after the restore completes.
- Update firewall/private-endpoint config for the new region.

Q107. How do you verify your backups are actually restorable rather than just present?

Answer:

- Regularly perform test restores to a scratch database and validate integrity.
- For VMs, run RESTORE VERIFYONLY and periodic full test restores.
- Automate a restore-and-DBCC-CHECKDB validation job on a schedule.
- Track restore success and duration to confirm RTO feasibility.

Q108. A stakeholder asks if they can restore over the existing production database in Azure SQL DB. What do you say?

Answer:

- PITR restores to a new database, not over the existing one.
- You restore to a new name, validate, then rename/repoint the app to swap.
- This preserves the original for rollback and comparison.
- Plan a brief cutover window for the rename/repoint step.

Q109. On a SQL VM, you need full control over backups. What's your strategy?

Answer:

- Use BACKUP TO URL to write native backups directly to Azure Blob Storage.
- Implement full/differential/log backups on a schedule (e.g., Ola Hallengren).
- Or use Azure Backup for SQL in VMs for managed, policy-driven backups.
- Monitor success/failure and test restores regularly.

Q110. Backup storage cost is climbing on Azure SQL. How do you control it?

Answer:

- Right-size PITR retention to actual needs (default 7 days; only extend when justified).
- Review LTR policies — long yearly retention accumulates significant storage.
- Consider locally- or zone-redundant backup storage instead of geo-redundant where residency allows.
- Monitor backup storage usage against the included allowance to avoid overage.

Q111. How do you recover from accidental data corruption caught days later?

Answer:

- If within PITR, restore to just before the corruption to a new database.
- If beyond PITR but within LTR, restore the nearest LTR full backup.
- Reconcile any transactions that occurred after the restore point.
- Investigate root cause (bad code, hardware) to prevent recurrence.

Q112. A team accidentally deleted an entire Azure SQL logical server. Is recovery possible?

Answer:

- Deleted servers can be recovered within a limited window, and databases via geo-restore of backups.
- Act quickly — restore the databases from their backups into a new/recovered server.
- Reconfigure firewall, Azure AD admin, auditing, and failover groups.
- Apply resource locks afterward to prevent recurrence.

Q113. Explain how backups work under the hood in Azure SQL Database.

Answer:

- The platform takes automatic full, differential, and frequent transaction-log backups.
- These enable second-level PITR without DBA scheduling.
- Backups are stored in read-access geo-redundant storage by default.
- You configure retention and LTR; the platform handles execution.

Q114. You need to hand a database to another team outside Azure. What portable format do you use?

Answer:

- Export a BACPAC (schema + data) and share it; they import into their SQL environment.
- For MI/VM, a native .bak backup is portable via RESTORE.
- BACPAC is logical and not transactionally consistent for a busy DB — export from a copy.
- Validate the import on the receiving side before decommissioning the source.

Q115. A restore succeeded but users report missing recent data. What happened and how do you prevent it?

Answer:

- The restore point was earlier than expected, or async replication lag lost recent transactions.
- Confirm the exact restore timestamp and reconcile the gap from other sources if possible.
- Tighten monitoring on restore-point selection and replication lag.
- For zero-loss needs, use synchronous HA rather than relying on backups alone.

Q116. How do you meet an aggressive RTO when the database is very large?

Answer:

- For very large DBs, use Hyperscale (near-instant restores) rather than classic tiers.
- For VMs, use compressed backups, fast storage, and parallel restore where possible.
- Pre-stage a warm standby (failover group/AG) so recovery is a failover, not a restore.
- Measure restore time in drills and size infrastructure to hit the RTO.

Q117. What's your policy for validating DBCC integrity in a PaaS world?

Answer:

- Azure runs automated integrity/consistency checks on managed backups behind the scenes.
- On MI/VM you still schedule DBCC CHECKDB on your databases.
- For Azure SQL Database, run CHECKDB on a restored copy if you need explicit verification.
- Alert on any consistency errors and open a support case if platform-managed checks fail.

Q118. A geo-restore is running but taking long. What determines the duration?

Answer:

- Database size, service tier of the target, and backup storage read throughput.
- Larger databases and lower target tiers restore slower.
- Restore into an adequately sized tier to speed it up, then rescale after.
- For frequent DR, prefer a running geo-secondary over geo-restore for lower RTO.

Q119. How do you back up and restore Always Encrypted or TDE-protected databases correctly?

Answer:

- TDE is transparent — backups are encrypted; ensure the target has access to the same key (Key Vault for BYOK).
- For BYOK/customer-managed keys, the restore target must reach the same Key Vault key.
- Always Encrypted column keys live in the client-side key store; ensure key access post-restore.
- Test the restore end-to-end including key access before relying on it.

Q120. You must prove to auditors that backups are geo-redundant. How?

Answer:

- Show the backup storage redundancy setting (geo-redundant) on the database.
- Demonstrate a successful geo-restore into the paired region.
- Provide LTR configuration evidence and retention duration.
- Include this in the compliance evidence pack with timestamps.

Q121. A costly LTR policy was set by mistake for 10 years on all databases. Remediation?

Answer:

- Review LTR policies per database and reduce to the compliance-required duration.
- Deleting future LTR policy stops new long-term backups; existing ones age out or can be removed per policy.
- Quantify the storage cost saved and document the corrected standard.
- Enforce LTR standards via governance to prevent recurrence.

Q122. How do you handle backups for an elastic pool with many databases?

Answer:

- Each database in the pool gets its own automated backups and PITR.
- Configure retention/LTR per database as needed.
- Backup storage is billed against the pool's allowance; monitor aggregate usage.
- Test restores of representative databases from the pool.

Q123. A restore must exclude a corrupt table but keep everything else. Approach?

Answer:

- Restore the full database to a new copy first.
- From the copy, export or copy out the good tables, or repair/replace the corrupt table.
- Selective object-level restore isn't native — you stage via a full copy.
- Validate referential integrity after reassembling the data.

Q124. What backup considerations change when using Hyperscale?

Answer:

- Hyperscale uses snapshot-based backups that are fast and size-independent.
- Restores and copies are much quicker than classic tiers.
- PITR retention is configurable; confirm it meets your operational needs.
- You still validate restores and configure geo-redundancy for DR.

Q125. How do you automate restore-testing across many databases?

Answer:

- Script periodic geo/PITR restores to scratch databases via PowerShell/CLI.
- Run integrity checks and record success/duration for RTO evidence.
- Schedule via Elastic Jobs, Automation runbooks, or a pipeline.
- Alert on any failure and drop the scratch databases after validation.

6. Performance Tuning & Query Optimization

DOMAIN 6

Performance Tuning & Query Optimization

30 real-time scenarios

Q126. Users report a query that was fast yesterday is slow today after no code change. What's your first tool in Azure SQL?

Answer:

- Open Query Store to compare the query's current plan against its historical fast plan.
- Identify a plan regression (parameter sniffing, stats change, CE change).
- Force the previously good plan via Query Store plan forcing to stabilize it.
- Then address root cause: update stats, add index, or fix the parameter-sensitive plan.

Q127. How does automatic tuning help you as an Azure SQL DBA?

Answer:

- It can auto-create beneficial indexes, drop duplicates, and force last-good plans on regression.
- Enable FORCE_LAST_GOOD_PLAN to auto-correct plan regressions.
- Review its recommendations and let it apply low-risk ones automatically.
- It reduces firefighting but you still validate index changes against write cost.

Q128. CPU is pinned at 100% on an Azure SQL Database. Walk through triage.

Answer:

- Check Query Store / sys.dm_exec_requests for top CPU-consuming queries.
- Look for missing indexes, bad plans, or excessive recompiles.
- Tune the offending queries/indexes before scaling compute.
- If the workload is legitimately heavy, scale up the tier or move to Serverless/Hyperscale.

Q129. A report query spills to tempdb and runs slowly. What do you investigate?

Answer:

- Check for underestimated row counts causing insufficient memory grants (spills).
- Update statistics and verify the cardinality estimate.
- Add supporting indexes to reduce sorts/hashes.
- Consider batch-mode/columnstore for large analytical scans.

Q130. You suspect parameter sniffing is causing inconsistent performance. Remedies?

Answer:

- Use OPTIMIZE FOR, RECOMPILE, or a plan guide for the specific query.
- Force a stable plan via Query Store if one plan is generally best.
- Consider parameter-sensitive plan optimization (newer engine feature) if available.
- Redesign to avoid one plan serving wildly different parameter ranges.

Q131. How do you find the most resource-intensive queries over the last week?

Answer:

- Query the Query Store views for top consumers by CPU, duration, logical reads, or memory.
- Sort by total or average metrics depending on whether frequency or per-execution cost matters.
- Correlate with wait statistics to see what they're waiting on.
- Prioritize tuning by aggregate impact, not single slow runs.

Q132. A DBA proposes adding many indexes to fix reads. What's the balanced view?

Answer:

- Indexes speed reads but slow writes and consume storage and maintenance overhead.
- Add only indexes justified by real query patterns and missing-index evidence.
- Consolidate overlapping indexes and drop unused ones.
- Measure write impact before finalizing on write-heavy tables.

Q133. Explain how you'd use wait statistics to diagnose a slow Azure SQL Database.

Answer:

- Query sys.dm_db_wait_stats (database-scoped in Azure SQL DB) for dominant wait types.
- PAGEIOLATCH points to I/O; LCK_ to blocking; SOS_SCHEDULER_YIELD to CPU; RESOURCE_SEMAPHORE to memory grants.
- Map the top waits to targeted fixes (indexing, isolation, scaling).
- Track waits over time to confirm the fix moved the bottleneck.

Q134. Blocking is causing timeouts. How do you find and fix the root blocker?

Answer:

- Use sys.dm_exec_requests / sys.dm_tran_locks (or Query Store + DMVs) to find the head blocker.
- Identify the long-running transaction holding locks.
- Fix by tuning the query, adding indexes, shortening transactions, or adjusting isolation.
- Consider RCSI/snapshot isolation to reduce reader-writer blocking.

Q135. A deadlock is reported intermittently. How do you capture and resolve it?

Answer:

- Capture the deadlock graph via Extended Events (system_health has deadlock captures).
- Analyze the resources and lock order to find the cycle.
- Fix by consistent object access order, better indexing, or shorter transactions.
- Add retry logic for deadlock victims as a safety net.

Q136. How do you decide whether to scale up versus tune queries when performance lags?

Answer:

- Tune first — scaling up masks inefficiency and raises cost permanently.
- If Query Store shows well-tuned queries still saturating resources, then scale.
- Use scaling for genuine growth; use tuning for regressions and bad plans.
- Right-size after tuning to avoid overpaying.

Q137. Intelligent Query Processing features exist in the engine. Which help you and how?

Answer:

- Batch mode on rowstore, memory grant feedback, and adaptive joins improve analytical/mixed workloads automatically.
- Table variable deferred compilation and interleaved execution improve estimates.
- Ensure the database compatibility level enables these features.
- Validate improvements via Query Store before/after.

Q138. A vendor app can't be changed but has a bad query. What non-code options do you have?

Answer:

- Force a good plan via Query Store without touching the query.
- Add a supporting index or updated statistics to change the optimizer's choice.
- Use plan guides to inject hints externally.

- Adjust compatibility level or database-scoped configurations if that stabilizes it.

Q139. How do you monitor query performance continuously rather than reactively?

Answer:

- Keep Query Store enabled with an appropriate capture mode and retention.
- Alert on regressions, high waits, and resource saturation via Azure Monitor.
- Baseline normal performance and compare periodically.
- Review top-resource queries weekly to catch drift early.

Q140. Statistics are stale on a large table and estimates are off. Approach?

Answer:

- Update statistics with an adequate sample (or fullscan) on the affected table.
- Enable auto-update stats; consider async auto-update to avoid blocking.
- For large tables, schedule targeted stats maintenance off-peak.
- Confirm the plan estimate improves in Query Store afterward.

Q141. A columnstore-eligible analytics query scans millions of rows slowly. What do you propose?

Answer:

- Add a clustered or nonclustered columnstore index for analytical scans.
- Columnstore + batch mode dramatically speeds aggregation over large data.
- Balance with OLTP needs — nonclustered columnstore on an OLTP table can serve both.
- Validate compression ratio and query improvement.

Q142. How do you tune tempdb-related contention in Azure SQL Managed Instance?

Answer:

- Ensure multiple tempdb data files (MI configures several by default).
- Reduce tempdb usage by fixing spills, large sorts, and excessive temp objects.
- Move to Business Critical for local-SSD tempdb latency if needed.
- Watch for PAGELATCH contention on tempdb allocation pages.

Q143. Read latency is high but CPU is fine. Where do you look in PaaS?

Answer:

- Check log I/O and data I/O percentage metrics — you may be I/O throttled by the tier.
- Business Critical (local SSD) offers lower latency than General Purpose (remote storage).
- Reduce I/O via indexing and query tuning to fit the tier.
- Scale the tier if I/O demand genuinely exceeds the limit.

Q144. A stored procedure recompiles constantly, wasting CPU. Causes and fixes?

Answer:

- Frequent stats changes, RECOMPILE hints, or SET option changes cause recompiles.
- Remove unnecessary RECOMPILE, stabilize stats, and standardize SET options.
- Consider optimizing for the common parameter set.
- Confirm recompile counts drop in Query Store.

Q145. How do memory grant issues manifest and how do you address them?

Answer:

- Excessive grants reduce concurrency (RESOURCE_SEMAPHORE waits); too-small grants cause spills.
- Fix estimates via updated stats and better indexes so grants are right-sized.
- Memory grant feedback (IQP) auto-corrects repeated mis-grants.
- Avoid hints that force large grants unless justified.

Q146. You need a repeatable performance baseline before a major release. How do you build it?

Answer:

- Capture Query Store metrics, wait stats, and resource utilization over a representative period.
- Record top queries and their plans/durations.
- After release, compare against the baseline to spot regressions.
- Automate baseline capture so it's consistent across releases.

Q147. A query uses a scalar UDF and is slow row-by-row. Modern remedy?

Answer:

- Scalar UDF inlining (IQP) can transform it into a set-based operation automatically at higher compat levels.
- Ensure the database compatibility level enables inlining.
- If not inlinable, rewrite the UDF logic inline or as a set-based join.
- Validate the plan and duration improvement afterward.

Q148. How do you handle a sudden workload spike gracefully in Azure SQL Database?

Answer:

- Serverless auto-scales compute within min/max bounds to absorb spikes.
- For provisioned tiers, pre-scale before known spikes or use automation to scale on metrics.
- Ensure app retry logic handles brief throttling during the ramp.
- Investigate whether the spike is efficient work or fixable inefficiency.

Q149. A partitioned table's queries aren't eliminating partitions. What do you check?

Answer:

- Verify the query predicate matches the partition column so partition elimination applies.
- Confirm the partition function/scheme is correct and aligned indexes exist.
- Check the plan for partition elimination in the operator properties.
- Adjust the query or partitioning strategy so scans are pruned.

Q150. How do you tell whether an index is helping or just adding overhead?

Answer:

- Check index usage stats (seeks/scans vs updates) via DMVs.
- High updates and near-zero seeks/scans indicate a costly, unused index.
- Cross-reference with Query Store to confirm no plan depends on it.
- Drop or consolidate unused/duplicate indexes after validation.

Q151. Reporting and OLTP contend on the same Azure SQL Database. Architectural fix?

Answer:

- Offload reporting to a read-scale replica (Business Critical/Premium) or a geo-secondary.
- Route ApplicationIntent=ReadOnly reads to the replica.
- For heavy analytics, consider a separate analytics store (Synapse/columnstore copy).
- This isolates workloads and protects OLTP latency.

Q152. How do you use extended events (or its Azure equivalent) for deep diagnosis?

Answer:

- Create lightweight XE sessions targeting the specific events (deadlocks, spills, long queries).
- In Azure SQL DB, use database-scoped XE with a suitable target (ring buffer/blob).
- Analyze captured data to pinpoint the exact failing statements.
- Keep sessions lightweight to avoid observer overhead in production.

Q153. A DBA wants to disable auto-update statistics to 'stabilize' plans. Your response?

Answer:

- Discourage it — stale stats cause worse estimates and unpredictable plans.
- Prefer async auto-update to avoid blocking on update.
- Use scheduled targeted stats maintenance for very large tables.
- Stabilize plans via Query Store forcing, not by disabling stats.

Q154. How do you validate that a tuning change actually improved things?

Answer:

- Compare before/after in Query Store for the target queries (duration, CPU, reads).
- Confirm the dominant wait type shifted or diminished.
- Check overall resource utilization didn't worsen elsewhere.
- Document the change and keep the ability to roll back (unforce plan, drop index).

Q155. Your database hits the max worker threads limit under load. What's happening and the fix?

Answer:

- Excessive concurrent requests (often from blocking) exhaust worker threads, causing THREADPOOL waits.
- Resolve the underlying blocking/long transactions to free workers.
- Reduce connection storms and add app-side pooling/retry.
- Scale the tier if concurrency legitimately exceeds the limit.

7. Security, Compliance & Data Protection

DOMAIN 7

Security, Compliance & Data Protection

30 real-time scenarios

Q156. Security mandates that data at rest must be encrypted for all Azure SQL databases. What do you confirm and configure?

Answer:

- Transparent Data Encryption (TDE) is enabled by default on Azure SQL Database and Managed Instance.
- For stricter control, use customer-managed keys (BYOK) stored in Azure Key Vault.
- Ensure the Key Vault key has soft-delete and purge protection enabled.
- Document key rotation and access policies for auditors.

Q157. How do you enforce that only corporate networks can reach an Azure SQL Database?

Answer:

- Disable public network access and use a Private Endpoint into your VNet.
- Traffic then flows over the Microsoft backbone, not the public internet.
- Configure Private DNS so the server FQDN resolves to the private IP.
- Remove permissive firewall rules like 0.0.0.0 and rely on private connectivity.

Q158. A team is using SQL logins everywhere. Why do you push Azure AD (Entra) authentication instead?

Answer:

- Entra auth centralizes identity, supports MFA and conditional access, and avoids password sprawl.
- Group-based access simplifies onboarding/offboarding and avoids orphaned users after failover.
- You can disable SQL authentication entirely for Entra-only enforcement.
- It integrates with PIM/just-in-time for privileged access.

Q159. Auditors require a record of who accessed sensitive tables. What do you enable?

Answer:

- Enable SQL Auditing to Log Analytics/Storage/Event Hub capturing the required action groups.
- Use SQL Data Discovery & Classification to label sensitive columns.
- Combine with auditing so access to classified data is logged.
- Set retention to meet the compliance mandate and alert on anomalous access.

Q160. How do you protect specific columns (e.g., SSNs) so even DBAs can't read them in clear text?

Answer:

- Use Always Encrypted so encryption/decryption happens client-side; the engine and DBAs see ciphertext.
- Store column master keys in a trusted key store (Key Vault or certificate store).
- Use deterministic encryption for equality lookups, randomized for maximum protection.
- Enclaves enable richer operations (range queries) on encrypted data where supported.

Q161. A report should show masked data to most users but full data to a few. What feature fits?

Answer:

- Dynamic Data Masking obscures data in query results for non-privileged users.
- Define masking rules per column (default, email, partial, random).
- Grant UNMASK to the few roles that need full values.
- Note it's a presentation-layer control, not a substitute for real encryption/authorization.

Q162. Financial-services compliance (SOX/PCI-DSS) is in scope. How do you approach the DBA controls?

Answer:

- Enforce least-privilege RBAC and SQL permissions with segregation of duties.
- Enable auditing of privileged actions, DDL, and access to cardholder/financial data.
- Encrypt data at rest (TDE/BYOK) and in transit (enforced TLS), plus column-level where required.
- Retain audit logs per mandate and produce evidence packs for auditors.

Q163. How do you enforce encryption in transit for Azure SQL connections?

Answer:

- Require Encrypt=True and validate the server certificate in connection strings.
- Set the minimum TLS version on the logical server/MI to 1.2 (or higher).
- Reject downgraded connections; monitor for legacy clients.
- Document the TLS policy for compliance.

Q164. Microsoft Defender for SQL is proposed. What does it give you as a DBA?

Answer:

- Vulnerability Assessment scans for misconfigurations with actionable remediation and baselines.
- Advanced Threat Protection alerts on SQL injection, anomalous logins, and data exfiltration patterns.
- It surfaces findings in Defender for Cloud with severity and guidance.
- Wire alerts to your SOC/action groups for response.

Q165. How do you manage privileged access to production databases safely?

Answer:

- Use Entra groups with Privileged Identity Management for just-in-time, time-boxed elevation.
- Require MFA and approval for privileged roles.
- Audit all privileged actions and review access periodically.
- Avoid standing admin access; grant least privilege by default.

Q166. A vulnerability assessment flags several findings. How do you triage and remediate?

Answer:

- Sort findings by severity and business risk.
- Remediate high-severity items (excess permissions, weak TLS, public access) first.
- Set an approved baseline for accepted-risk items so recurring scans are clean.
- Track remediation to closure and re-scan to confirm.

Q167. How do you rotate customer-managed TDE keys without downtime?

Answer:

- Add a new key version in Key Vault and update the TDE protector to the new version.
- TDE re-encrypts the database encryption key online; data stays available.
- Keep old key versions until re-encryption completes to avoid access loss.
- Document rotation cadence to satisfy compliance.

Q168. A subnet must never allow databases with public access. How do you guarantee it org-wide?

Answer:

- Use Azure Policy to deny SQL servers/MIs with public network access enabled.
- Enforce private endpoints via policy/initiative.
- Assign at management-group scope for consistency.
- Audit existing resources and remediate non-compliant ones.

Q169. How do you prevent SQL injection risks from a DBA standpoint?

Answer:

- Enforce parameterized queries/stored procedures at the app layer (advise developers).
- Enable Defender for SQL to detect injection attempts at runtime.
- Apply least privilege so a compromised account can't do broad damage.
- Monitor audit logs for suspicious query patterns.

Q170. Sensitive data must be discoverable for a data-governance program. What do you use?

Answer:

- SQL Data Discovery & Classification to label columns (e.g., PII, financial).
- Export the classification report for the governance catalog.
- Integrate with Microsoft Purview for enterprise-wide data cataloging and lineage.
- Combine labels with auditing and masking for end-to-end protection.

Q171. How do you securely store and use application connection secrets?

Answer:

- Store connection strings/secrets in Azure Key Vault, not in code or config files.
- Prefer managed identities so apps authenticate to SQL without secrets at all.
- Grant the app's managed identity a scoped SQL role.
- Rotate any residual secrets and audit access to the vault.

Q172. A managed identity should access Azure SQL instead of a password. How do you set it up?

Answer:

- Enable a system- or user-assigned managed identity on the app resource.
- Create a contained database user FROM EXTERNAL PROVIDER for that identity.
- Grant it the minimum required role.
- Update the connection to use Entra token auth (no password).

Q173. How do you handle the principle of least privilege for DBAs versus developers?

Answer:

- Grant DBAs operational roles; grant developers read-only or scoped app roles.
- Avoid db_owner/sysadmin unless strictly required and time-boxed.
- Use custom database roles mapped to Entra groups.
- Review and recertify access regularly.

Q174. An auditor asks how you'd detect and respond to a data exfiltration attempt.

Answer:

- Defender for SQL threat alerts flag anomalous export/access patterns.
- Auditing captures the queries and principals involved.
- Alert routing notifies the SOC for immediate investigation.
- Have an incident runbook: contain, investigate, rotate credentials, report.

Q175. How do you meet data-residency requirements strictly?

Answer:

- Provision only in approved regions and restrict others via Azure Policy.
- Use locally- or zone-redundant backup storage to avoid paired-region replication.
- Disable geo-replication if residency forbids cross-border copies.
- Document the residency posture for compliance evidence.

Q176. A shared logical server hosts multiple apps. How do you isolate their security?

Answer:

- Separate databases with distinct users/roles; avoid cross-database access unless required.
- Prefer separate logical servers/MIs for strong isolation of sensitive workloads.
- Apply per-database auditing and least-privilege roles.
- Use private endpoints and network segmentation per app tier.

Q177. How do you enforce MFA for human access to Azure SQL?

Answer:

- Use Entra authentication with Conditional Access requiring MFA.
- Restrict interactive access to Entra identities, not SQL logins.
- Combine with PIM for privileged, time-boxed access.
- Audit sign-ins via Entra logs.

Q178. What's your approach to certificate and key lifecycle for encryption features?

Answer:

- Centralize keys/certs in Key Vault with soft-delete and purge protection.
- Automate rotation and set expiry alerts.
- Restrict key access via RBAC/access policies to a minimal set.
- Test recovery: ensure restores can reach the required keys.

Q179. How do you respond when a former employee's access wasn't revoked?

Answer:

- If using Entra groups, removing them from the group revokes access immediately.
- Audit what they accessed while active via SQL audit logs.
- Rotate any shared secrets they may have known.
- Improve the offboarding process to be group/identity-driven and automated.

Q180. A compliance framework requires evidence of encryption everywhere. What do you compile?

Answer:

- TDE status (and BYOK key details) per database.
- Enforced TLS version and Encrypt=True policy.
- Column-level Always Encrypted coverage for sensitive fields.
- Backup encryption and key access documentation.

Q181. How do you secure a hybrid connection between on-prem and Azure SQL/MI?

Answer:

- Use ExpressRoute or site-to-site VPN for private connectivity.
- Use private endpoints/MI subnet so traffic avoids the public internet.
- Enforce TLS and Entra auth end to end.
- Segment networks with NSGs and monitor the link.

Q182. Ledger is mentioned for tamper-evidence. When would you use it?

Answer:

- Use SQL Ledger to provide cryptographic tamper-evidence for sensitive tables.
- Updatable or append-only ledger tables record verifiable history.
- It helps with audit/compliance where data integrity must be provable.
- Combine with auditing for a complete trust story.

Q183. How do you handle least-privilege for automation service principals?

Answer:

- Grant the SPN/managed identity only the specific RBAC and SQL roles it needs.
- Scope to the exact resource group/database, not the subscription.
- Prefer managed identities over client secrets; rotate any secrets used.
- Audit the automation identity's actions.

Q184. A pen test recommends disabling legacy protocols. What do you check on Azure SQL?

Answer:

- Set minimum TLS to 1.2+ on the server/MI and reject older versions.
- Disable SQL authentication in favor of Entra-only if feasible.
- Remove overly-broad firewall rules and enforce private endpoints.
- Re-run the assessment to confirm the findings are closed.

Q185. How do you demonstrate segregation of duties for SOX compliance?

Answer:

- Separate roles: those who deploy changes cannot approve them and vice versa.
- Use RBAC/PIM so privileged access is requested, approved, and audited.
- Audit all privileged and DDL actions with immutable retention.
- Produce access-review evidence showing periodic recertification.

8. Monitoring, Alerting & Diagnostics

DOMAIN 8

Monitoring, Alerting & Diagnostics

25 real-time scenarios

Q186. You need a single pane of glass for all Azure SQL resources. How do you build it?

Answer:

- Send diagnostic settings (metrics + logs) to a Log Analytics workspace.
- Build Azure Monitor Workbooks/dashboards querying the data with KQL.
- Add alert rules on key metrics (CPU, DTU/vCore, storage, deadlocks, failed connections).
- Optionally use Azure SQL Analytics/database watcher for richer fleet insights.

Q187. Which core metrics do you alert on for an Azure SQL Database?

Answer:

- CPU percentage, DTU/vCore utilization, data and log I/O percentage.
- Storage used percentage and worker/session percentage.
- Deadlocks, failed connections, and blocked-by-firewall connections.
- For Serverless, track auto-pause status and app-caused constant connections.

Q188. How do you get query-level telemetry into your monitoring platform?

Answer:

- Enable Query Store and route the relevant categories to Log Analytics via diagnostic settings.
- Capture QueryStoreRuntimeStatistics/WaitStatistics logs for KQL analysis.
- Build workbooks for top consumers and regressions.
- Alert on regression signals where feasible.

Q189. An alert fires for high CPU at 2 AM. What should the alert contain to be actionable?

Answer:

- The resource, metric, threshold breached, and current value.
- A link to the dashboard/workbook and the runbook to follow.
- Context: top queries if enriched, and recent scaling/deployment events.
- Routing to the correct on-call via action groups (SMS/email/webhook/ITSM).

Q190. How do you avoid alert fatigue while still catching real issues?

Answer:

- Use dynamic thresholds and appropriate aggregation windows to reduce noise.
- Set severity levels and route only actionable alerts to on-call.
- Suppress alerts during known maintenance windows.
- Review and tune alert rules regularly based on signal-to-noise.

Q191. You must prove SLA/availability to leadership monthly. How do you report it?

Answer:

- Track availability and failed-connection metrics in Log Analytics.
- Build a monthly workbook showing uptime, incidents, and RTO/RPO adherence.
- Include failover events and their impact.
- Automate the report generation on a schedule.

Q192. How do you detect storage nearing capacity before it causes an outage?

Answer:

- Alert on storage-used-percentage crossing a warning threshold (e.g., 80%).
- For MI/VM, monitor per-file/log growth too.
- Automate remediation (scale storage) or notify for manual action.
- Trend growth to plan capacity ahead of thresholds.

Q193. A workbook shows periodic latency spikes. How do you correlate the cause?

Answer:

- Overlay resource metrics with Query Store waits and deployment/scaling events on the same timeline.
- Check for scheduled jobs, batch loads, or maintenance windows aligning with spikes.
- Drill into top queries during spike windows.
- Confirm the fix by watching the spike disappear post-change.

Q194. How do you monitor a fleet of many databases efficiently?

Answer:

- Centralize diagnostics into one Log Analytics workspace across resources.
- Use database watcher or Azure SQL Analytics for consolidated fleet views.
- Tag resources for scoping alerts and dashboards by environment/app.
- Create fleet-level KQL queries to find outliers automatically.

Q195. What's your approach to log retention for diagnostics and audit data?

Answer:

- Set Log Analytics retention to meet operational needs, and archive long-term to storage.
- Separate audit logs (often longer, compliance-driven retention) from operational logs.
- Balance retention cost against investigation and compliance requirements.
- Document the retention policy.

Q196. How do you alert on deadlocks and act on them?

Answer:

- Enable the deadlock metric/log and alert when counts spike.
- Capture deadlock graphs via Extended Events (system_health) for root cause.
- Fix via indexing, access order, and shorter transactions.
- Track whether the deadlock rate drops after the fix.

Q197. A stakeholder wants proactive rather than reactive DBA operations. What do you implement?

Answer:

- Baselines + trend-based alerts to catch drift before incidents.
- Automated remediation runbooks for common conditions (scale, restart jobs).
- Regular health reviews of top queries, waits, and capacity.
- Post-incident reviews feeding back into alert/runbook improvements.

Q198. How do you monitor replication/failover health for HA/DR?

Answer:

- Alert on replication lag, replica health state, and failover-group link health.
- For VMs, monitor AG/WSFC health and synchronization state.
- Set thresholds tied to your RPO to warn before breach.
- Include HA/DR panels in the operational dashboard.

Q199. You need to capture a rare, intermittent error for RCA. Tooling?

Answer:

- Set up a targeted Extended Events session for the specific error/event.
- Route the target to blob/ring buffer and keep it lightweight.
- Correlate captures with Query Store and metrics at the failure time.
- Remove the session once the root cause is found.

Q200. How do you integrate Azure SQL alerts with your ITSM/on-call tooling?

Answer:

- Use Action Groups with webhook/ITSM connectors (e.g., ServiceNow) and paging integrations.
- Map severity to routing and escalation policies.
- Include runbook links in the payload for responder context.
- Test the end-to-end alert-to-ticket flow.

Q201. A Serverless database stopped auto-pausing, inflating cost. How do you detect and fix it?

Answer:

- Monitor the pause status metric; constant connections prevent pausing.
- Identify the app/monitoring tool holding connections open.
- Adjust the app to release idle connections or increase auto-pause delay appropriately.
- Confirm pausing resumes and cost drops.

Q202. How do you baseline 'normal' so anomalies stand out?

Answer:

- Collect metrics/query stats over a representative period (including peaks and troughs).
- Store baseline values and use dynamic thresholds keyed off them.
- Re-baseline after major workload/release changes.
- Alert on deviations from the baseline, not just static numbers.

Q203. What KQL-based checks do you run to find problem databases quickly?

Answer:

- Query for databases with sustained high CPU/IO/storage percentages.
- Find top failed-connection and deadlock counts across the fleet.
- List Query Store regressions and long-running queries.
- Rank by impact to prioritize remediation.

Q204. How do you monitor SQL on Azure VMs differently from PaaS?

Answer:

- Install the Azure Monitor agent and enable VM insights plus SQL best-practices assessment.
- Collect Windows/SQL perf counters, error logs, and AG health into Log Analytics.
- Monitor disk latency/IOPS at the VM level (which PaaS abstracts).
- Alert on OS-level issues (memory, disk) as well as SQL.

Q205. An executive wants a cost-and-performance overview together. How do you present it?

Answer:

- Combine Cost Management data with performance metrics in a single workbook.
- Show utilization vs spend to highlight over/under-provisioning.
- Recommend right-sizing, reserved capacity, or AHB where justified.
- Refresh monthly for trend visibility.

Q206. How do you validate that monitoring itself is working (no blind spots)?

Answer:

- Periodically trigger a test alert to confirm the pipeline delivers.
- Check that diagnostic settings are enabled on every resource (enforce via Policy).
- Verify Log Analytics is receiving data from all sources.
- Review for resources missing coverage after new deployments.

Q207. A database shows high log I/O. What does that indicate and how do you monitor it?

Answer:

- High log write rate often means heavy DML/transaction volume or large batch operations.
- Alert on log I/O percentage nearing the tier limit.
- Tune by batching, reducing per-row transactions, and minimizing logging where safe.
- Scale the tier if the write throughput is legitimately required.

Q208. How do you track long-term capacity trends for planning?

Answer:

- Retain metrics long enough (or archive) to see multi-month trends.
- Build trend charts for CPU, storage, IOPS, and connections.
- Project against business growth to time scaling/reservations.
- Review quarterly with stakeholders.

Q209. What's your approach to auditing changes made by DBAs and automation?

Answer:

- Enable Azure Activity Log and SQL Audit to capture control-plane and data-plane changes.
- Route to Log Analytics with alerting on sensitive changes.
- Correlate who/what/when for accountability.
- Retain per compliance and review periodically.

Q210. How do you set up alerting for backup failures across the estate?

Answer:

- For VMs/Azure Backup, alert on backup job failures via the vault's monitoring.
- For PaaS, monitor restore-test automation results and LTR job health.
- Route failures to on-call with runbook links.
- Track backup success rate as an SLO.

9. Migration & Modernization

DOMAIN 9

Migration & Modernization

25 real-time scenarios

Q211. You must migrate 50 on-prem SQL databases to Azure with minimal downtime. What's your end-to-end approach?

Answer:

- Assess with Data Migration Assistant / Azure Migrate for compatibility and target-tier recommendations.
- Choose the target (Azure SQL DB, MI, or VM) per feature and compatibility needs.
- Use Azure Database Migration Service (online mode) or MI Log Replay Service to minimize downtime.
- Cut over during a short window after continuous sync, then validate and decommission.

Q212. How do you assess whether an on-prem database can go to Azure SQL Database vs Managed Instance?

Answer:

- Run Data Migration Assistant to find blocking/behavioral compatibility issues.
- Instance-scoped features (SQL Agent, cross-DB, CLR, linked servers) push you to MI.
- Single-database, elastic-pool workloads without those features fit Azure SQL Database.
- Document required remediations before committing to a target.

Q213. What's the difference between offline and online migration, and when do you choose each?

Answer:

- Offline: one-time restore/import with downtime for the whole migration.
- Online: continuous sync (log replay/CDC) with a short cutover, minimizing downtime.
- Choose offline for small/low-criticality DBs and generous windows.
- Choose online for large, business-critical databases needing near-zero downtime.

Q214. How do you migrate to Managed Instance with the Log Replay Service?

Answer:

- Take full, then differential and log backups of the source to Azure Blob Storage.
- Start LRS to restore and continuously replay logs onto the target MI.
- Keep replaying until cutover, then complete the migration to finalize.
- Validate data, logins, and jobs before repointing the app.

Q215. A stakeholder wants to lift-and-shift a whole SQL Server with the OS. Which target and tool?

Answer:

- Target SQL Server on Azure VM (IaaS) for full OS/SQL control.
- Use Azure Migrate for server-level replication and cutover.
- Post-migration, register the SQL IaaS extension and configure backups/HA.
- Validate performance and licensing (apply AHB).

Q216. How do you handle logins and orphaned users during migration?

Answer:

- Script and recreate server-level logins on the target (or use Entra groups).
- Fix orphaned database users by remapping SIDs or using contained users.
- For failover groups, ensure logins exist on both instances.
- Validate access with representative accounts post-migration.

Q217. Post-migration, some queries regressed. Why and how do you stabilize them?

Answer:

- New cardinality estimator/compat level can change plans.
- Use Query Store to compare and force previously good plans.
- Update statistics and validate indexes migrated correctly.
- Adjust compatibility level cautiously if needed, testing thoroughly.

Q218. How do you validate a migration before final cutover?

Answer:

- Compare schema, object counts, and row counts source vs target.
- Run application smoke tests and key business queries.
- Validate jobs, linked servers, permissions, and performance vs baseline.
- Confirm backups, HA/DR, and monitoring are configured on the target.

Q219. A large database's initial seeding over the network is too slow. Options?

Answer:

- Use compressed backups and restore-from-URL to speed transfer.
- For huge datasets, consider Azure Data Box for offline seeding.
- Then apply differential/log restores to catch up before cutover.
- Parallelize where the tooling supports it.

Q220. How do you minimize cutover downtime for a critical database?

Answer:

- Use online migration with continuous log replay/CDC so only the final delta applies at cutover.
- Pre-stage everything (logins, jobs, config) on the target beforehand.
- Cut over in a short window: stop writes, apply final logs, repoint app.
- Have a rollback plan if validation fails.

Q221. You're migrating SSIS/ETL alongside databases. What's the target?

Answer:

- Rehost SSIS packages on the SSIS Integration Runtime in Azure Data Factory.
- Or refactor to ADF/Synapse pipelines for cloud-native ETL.
- Validate connections, drivers, and package dependencies.
- Schedule and monitor the migrated packages in ADF.

Q222. How do you migrate SSRS reports to Azure?

Answer:

- Host SSRS on a SQL VM or use Power BI Report Server / Power BI paginated reports.
- Migrate report definitions and data sources, repointing to Azure databases.
- Validate rendering, subscriptions, and security.
- Decommission the old report server after validation.

Q223. A migration must handle a version upgrade (e.g., 2016 to current) too. Approach?

Answer:

- Assess deprecated features and behavior changes for the target version.
- Migrate to a target running the newer engine (MI is evergreen; VM/DB you choose).
- Test at the new compatibility level and address plan/behavior changes.
- Roll out gradually with Query Store as a safety net.

Q224. How do you choose between DMS, native backup/restore, and replication for a migration?

Answer:

- DMS: managed, supports online modes and many sources.
- Native backup/restore (to URL / to MI): simple for MI/VM targets, offline-oriented.
- Transactional replication/CDC: near-zero-downtime for specific scenarios.
- Pick based on target, size, downtime tolerance, and source features.

Q225. A stakeholder asks how to de-risk a big-bang migration. What do you recommend?

Answer:

- Migrate in waves by application/criticality rather than all at once.
- Pilot with a low-risk database to prove the runbook.
- Keep rollback options (source intact until validated).
- Communicate schedules and success criteria clearly.

Q226. How do you handle collation or feature incompatibilities discovered in assessment?

Answer:

- Match target collation to source; MI lets you set instance collation at creation.
- Remediate unsupported features (e.g., replace deprecated syntax) before migration.
- Test the fixes in a staging target.
- Document any behavioral differences for the app team.

Q227. You need to migrate with the app team unable to change connection strings easily. Idea?

Answer:

- Use a stable endpoint (failover-group listener) and DNS aliasing to minimize app changes.
- Point the existing DNS name to the new endpoint at cutover.
- Coordinate the DNS TTL so the switch propagates quickly.
- Provide the app team retry-logic guidance for the transition.

Q228. How do you validate performance parity after migration to PaaS?

Answer:

- Compare Query Store metrics and top queries against the on-prem baseline.
- Load test representative workloads on the target tier.
- Right-size the tier based on observed utilization.
- Address regressions before declaring success.

Q229. A migration wave failed validation. What's your rollback discipline?

Answer:

- Keep the source authoritative until the target passes validation.
- If validation fails, repoint the app back to the source (no data written to target yet in online mode pre-cutover).
- Capture the failure cause, fix, and reschedule.
- Never decommission the source prematurely.

Q230. How do you migrate a database to Hyperscale and what changes?

Answer:

- Convert/migrate into the Hyperscale tier for very large or fast-growing databases.
- Backups become snapshot-based (fast); scaling compute is quick.
- Add read replicas for scale-out reads.
- Note Hyperscale has specific behaviors — validate features your app relies on.

Q231. What's your approach to migrating an entire estate cost-effectively?

Answer:

- Assess and right-size targets from real utilization, not peak-of-peak.
- Apply Azure Hybrid Benefit and reserved capacity where steady.
- Consolidate small databases into elastic pools/MI where isolation allows.
- Track cost post-migration and optimize iteratively.

Q232. How do you keep source and target in sync during a long parallel-run period?

Answer:

- Use online migration/CDC/log replay to continuously apply changes.
- Monitor replication lag and reconcile row counts periodically.
- Freeze schema changes on the source during the sync window.
- Cut over once lag is minimal and validation passes.

Q233. A team wants to modernize from IaaS SQL to PaaS later. How do you plan the path?

Answer:

- Start on VM for lift-and-shift, then assess PaaS compatibility over time.
- Remove blockers (unsupported features) incrementally.
- Migrate to MI (high compatibility) or Azure SQL DB when ready.
- Use the same online-migration tooling for the second hop.

Q234. How do you migrate while meeting strict compliance during the process?

Answer:

- Encrypt data in transit for all migration traffic (TLS, private connectivity).
- Keep migration within approved regions for residency.
- Audit the migration actions and restrict access to the tooling identities.
- Validate encryption/auditing is active on the target before go-live.

Q235. What post-migration hardening steps do you always perform?

Answer:

- Configure backups, HA/DR, and monitoring/alerting on the target.
- Apply security baseline: TDE/BYOK, private endpoints, Entra auth, auditing.
- Right-size the tier and apply AHB/reservations.
- Update runbooks and decommission the source after a stabilization period.

10. Automation, PowerShell, CLI & Job Orchestration

DOMAIN 10

Automation, PowerShell, CLI & Job Orchestration

20 real-time scenarios

Q236. You must run index maintenance across 100 Azure SQL databases on a schedule. What's your approach?

Answer:

- Use Elastic Jobs to run T-SQL maintenance across many databases in parallel from a job database.
- Define target groups (server/pool/database) and a schedule.
- Parameterize the maintenance script (e.g., reorganize/rebuild by fragmentation).
- Monitor job execution history and alert on failures.

Q237. How do you automate scaling a database up before nightly batch and down after?

Answer:

- Use an Azure Automation runbook or Logic App triggered on schedule to call the scale API.
- Scale up before the batch window, scale down after completion.
- Guard with checks so you don't scale during active failovers.
- Log the actions and alert on failures.

Q238. A repeatable database provisioning process is needed. Which tooling and why?

Answer:

- Use Bicep/ARM or Terraform for the resources and SqlPackage/DACPAC for schema.
- Drive it through a CI/CD pipeline (Azure DevOps/GitHub Actions).
- Parameterize environment differences (tier, region, firewall).
- Version everything in Git for auditability and rollback.

Q239. You want to standardize DBA scripts across environments. How do you manage them?

Answer:

- Store scripts in a Git repo with clear structure and reviews.
- Use PowerShell modules (including dbatools where applicable) for reusable functions.
- Parameterize connections and secrets via Key Vault/managed identity.
- Run through pipelines rather than ad-hoc on servers.

Q240. How do you use dbatools in an Azure context?

Answer:

- dbatools works against SQL on Azure VM and Managed Instance for many operations (backups, migrations, checks).
- For Azure SQL Database, some cmdlets apply; combine with Az PowerShell for platform actions.
- Use it to automate assessments, copies, and maintenance tasks.
- Secure credentials via Key Vault/managed identity, not plaintext.

Q241. A team keeps making manual portal changes causing drift. How do you stop it?

Answer:

- Move to infrastructure-as-code so the desired state is declared and enforced.
- Use Azure Policy to audit/deny non-compliant configurations.
- Restrict portal write access via RBAC; changes go through pipelines.
- Detect drift by re-running IaC in what-if/plan mode.

Q242. How do you schedule cross-database T-SQL that needs to run everywhere consistently?

Answer:

- Elastic Jobs for Azure SQL Database targets across servers/pools.
- SQL Agent for MI/VM targets.
- Central Management Server pattern for VM fleets.
- Alert on failures and centralize execution history.

Q243. You need to rotate credentials/keys automatically. How?

Answer:

- Store secrets/keys in Key Vault and enable rotation policies/events.
- Prefer managed identities to remove long-lived secrets entirely.
- Automate any residual rotation via runbooks reacting to Key Vault events.
- Audit access and confirm apps pick up new secrets.

Q244. How do you automate backup-restore testing for RTO validation?

Answer:

- Script periodic PITR/geo restores to scratch databases via Az PowerShell/CLI.
- Run integrity checks and record duration for RTO evidence.
- Schedule via Automation/Elastic Jobs/pipeline.
- Alert on failures and clean up scratch databases.

Q245. A runbook must react to an alert (e.g., high storage). How do you wire it?

Answer:

- Configure an Action Group to trigger the Automation runbook/Logic App webhook on the alert.
- The runbook performs remediation (scale storage) idempotently.
- Log actions and notify humans of what was done.
- Add guardrails to avoid runaway automation.

Q246. How do you manage secrets for automation without hardcoding them?

Answer:

- Use managed identities for Azure resources to authenticate without secrets.
- Store any required secrets in Key Vault and reference them at runtime.
- Scope the identity's RBAC/SQL permissions minimally.
- Never commit secrets to source control.

Q247. You must deploy schema changes safely and repeatably. What's the pattern?

Answer:

- Use DACPAC (SqlPackage) or migration-based tooling (EF migrations/Flyway) in CI/CD.
- Review changes via pull requests and deploy through pipelines.
- Include pre/post-deployment scripts and rollback strategy.
- Validate in lower environments before production.

Q248. How do you automate consistent tagging and governance for new databases?

Answer:

- Enforce required tags via Azure Policy with deny/modify effects.
- Bake tags into IaC modules so deployments comply by default.
- Backfill existing resources with a script.
- Use tags to scope automation (e.g., pause dev nightly).

Q249. A DBA wants to pause all dev databases nightly to save cost. Implementation?

Answer:

- Move dev to Serverless with auto-pause, or automate scale/stop via runbook on schedule.
- Target dev resources by tag.
- Skip anything mid-deployment or in use.
- Resume before business hours and verify state.

Q250. How do you keep automation idempotent and safe?

Answer:

- Design runbooks/IaC to converge to a desired state regardless of current state.
- Add checks before destructive actions and dry-run/what-if modes.
- Log every action and make operations retryable.
- Test automation in non-prod before prod.

Q251. You need to bulk-configure auditing/diagnostics on many servers. Approach?

Answer:

- Use Azure Policy with deployIfNotExists to auto-enable diagnostic settings/auditing.
- Or script it across resources via Az PowerShell/CLI.
- Verify coverage and remediate gaps.
- Enforce for new resources automatically.

Q252. How do you orchestrate a multi-step DR failover with automation?

Answer:

- Script the failover (failover group/AG), DNS updates, and validation steps in a runbook.
- Include health checks and app smoke tests as gates.
- Make it re-runnable and log each step.
- Rehearse it in DR drills so it's proven.

Q253. A pipeline must deploy to dev, test, and prod with approvals. How do you structure it?

Answer:

- Use environment stages with gated approvals in the CI/CD tool.
- Parameterize per-environment values; keep secrets in Key Vault.
- Run automated validation between stages.
- Require sign-off for prod (segregation of duties).

Q254. How do you centrally run health checks across a SQL VM fleet?

Answer:

- Use a Central Management Server + registered servers to run scripts fleet-wide.
- Or Azure Automation with a runbook iterating the inventory.
- Aggregate results into Log Analytics for dashboards.
- Alert on failed checks.

Q255. What's your philosophy on automation vs manual DBA work?

Answer:

- Automate repetitive, error-prone, and scale-out tasks; keep humans for judgment calls.
- Ensure automation is observable, idempotent, and reversible.
- Document runbooks so automation is understood, not a black box.
- Continuously improve based on incidents and toil reduction.

11. Elastic Pools, Scaling & Hyperscale

DOMAIN 11

Elastic Pools, Scaling & Hyperscale

20 real-time scenarios

Q256. 40 SaaS-tenant databases have bursty, non-overlapping load. Why use an elastic pool?

Answer:

- Pools share compute across databases, smoothing cost when peaks don't coincide.
- You pay for pool capacity rather than sizing each database for its own peak.
- Per-database min/max caps prevent one tenant starving others.
- Add/move databases in/out of the pool as demand changes.

Q257. How do you decide the right size for an elastic pool?

Answer:

- Estimate aggregate peak demand across databases, accounting for overlap probability.
- Set eDTU/vCore pool capacity to cover realistic concurrent peaks.
- Configure per-database min to guarantee baseline, max to cap greedy tenants.
- Monitor pool utilization and resize as the tenant mix grows.

Q258. A single tenant in a pool is consistently saturating resources. What do you do?

Answer:

- Cap its per-database max within the pool to protect neighbors.
- If it needs more, move it to its own database/tier.
- Investigate whether its workload can be tuned.
- Re-balance the pool composition based on utilization data.

Q259. How do you scale an elastic pool without downtime?

Answer:

- Resize the pool's eDTU/vCore online; a brief reconnect may occur at switch-over.
- Automate resizing on a schedule or metric via runbook.
- Ensure app retry logic handles the brief transition.
- Validate per-database caps still make sense after resizing.

Q260. When do you split one elastic pool into multiple pools?

Answer:

- When aggregate demand outgrows a single pool's ceiling.
- When you need to isolate workloads (e.g., by environment or SLA).
- When per-database caps can't reconcile very different tenant profiles.
- Group databases with similar patterns into the same pool.

Q261. A database in a pool needs its own SLA/tier. How do you handle it?

Answer:

- Move it out of the pool into a standalone database at the required tier.
- This gives dedicated resources and independent scaling.
- Update monitoring/cost tracking for the standalone resource.
- Keep the pool for the remaining shared workloads.

Q262. What is Hyperscale and when do you choose it?

Answer:

- A tier that decouples compute, log service, and page servers for huge, fast-growing databases.
- Supports very large sizes (100+ TB), near-instant backups, and fast scaling.
- Add read replicas for scale-out reads.
- Choose it for large OLTP/mixed workloads outgrowing classic tiers.

Q263. How does Hyperscale change backup and restore behavior?

Answer:

- Backups are snapshot-based, fast, and size-independent.
- Restores and database copies are much quicker than classic tiers.
- PITR is available; confirm retention meets needs.
- This reduces RTO for very large databases dramatically.

Q264. A Hyperscale database has heavy reporting reads. How do you scale reads?

Answer:

- Add named/HA read replicas and route read-only connections to them.
- Replicas scale independently of the primary.
- Use ApplicationIntent/endpoints to direct reporting traffic.
- Monitor replica lag for reporting freshness.

Q265. How do you scale compute quickly for a sudden sustained load increase?

Answer:

- Serverless auto-scales within min/max bounds automatically.
- For provisioned tiers, scale vCores online (brief reconnect at the end).
- Hyperscale scales compute rapidly due to decoupled storage.
- Pre-scale for predictable spikes; automate for reactive ones.

Q266. A Serverless database's cost is unpredictable. How do you make it more predictable?

Answer:

- Set min/max vCores to bound compute cost.
- Tune auto-pause delay to balance responsiveness and savings.
- If load is steady, provisioned + reserved capacity may be cheaper.
- Monitor compute-seconds billed to understand the pattern.

Q267. How do you handle scaling storage independently of compute?

Answer:

- vCore model lets you grow storage without changing compute (within tier limits).
- Hyperscale grows storage automatically as needed.
- Monitor storage-used and alert before limits.
- Right-size compute separately based on CPU/IO.

Q268. A pool's per-database min settings sum to more than the pool capacity. What happens?

Answer:

- The platform won't guarantee mins that exceed pool capacity — you must keep sum(min) within capacity.
- Adjust mins so guarantees are feasible.
- Increase pool size if all databases genuinely need the baseline.
- Use monitoring to right-size mins to real usage.

Q269. How do you migrate a standalone database into an elastic pool?

Answer:

- Move the database into the pool (online) via portal/PowerShell/CLI.
- Set appropriate per-database min/max within the pool.
- Validate performance under shared capacity.
- Monitor pool utilization after adding it.

Q270. When is Hyperscale not the right choice?

Answer:

- For small databases where classic tiers are cheaper and simpler.
- When a specific feature your app needs behaves differently on Hyperscale — validate first.
- When you require certain classic-tier behaviors not present in Hyperscale.
- Assess features and cost before converting.

Q271. How do you plan capacity for a growing multi-tenant pool over a year?

Answer:

- Trend aggregate pool utilization and tenant growth.
- Model when you'll need to resize or split pools.
- Reserve capacity once steady-state is known.
- Keep per-database caps aligned with tenant SLAs.

Q272. A tenant needs isolation for compliance but currently shares a pool. Fix?

Answer:

- Move that tenant to a dedicated database (or separate server/MI) for isolation.
- Apply tenant-specific security and auditing.
- Keep other tenants in the shared pool.
- Document the isolation for the compliance record.

Q273. How do read replicas differ between Business Critical and Hyperscale?

Answer:

- Business Critical provides one free readable secondary via read scale-out.
- Hyperscale supports multiple named read replicas that scale independently.
- Choose Hyperscale when you need several read replicas for scale.
- Route read-only traffic accordingly in both cases.

Q274. A database frequently hits its storage cap in a classic tier. Long-term fix?

Answer:

- If growth is sustained, move to Hyperscale for large, auto-growing storage.
- Otherwise increase storage within the vCore tier limit.
- Archive/purge old data if retention allows.
- Alert on storage trends to plan ahead.

Q275. How do you validate that pooling actually saves money for your estate?

Answer:

- Compare the sum of standalone tier costs vs the pool cost at required capacity.
- Use utilization data to confirm peaks don't fully overlap.
- Recompute periodically as the tenant mix changes.
- Split or consolidate pools based on the analysis.

12. Networking & Connectivity

DOMAIN 12

Networking & Connectivity

15 real-time scenarios

Q276. An app must reach Azure SQL Database privately with no public exposure. How do you architect it?

Answer:

- Create a Private Endpoint for the logical server in the app's VNet subnet.
- Disable public network access on the server.
- Configure Private DNS zone so the FQDN resolves to the private IP.
- Validate the app connects over the private path only.

Q277. What's the difference between service endpoints and private endpoints for Azure SQL?

Answer:

- Service endpoints extend the VNet identity to the service over the Azure backbone but the resource keeps a public IP.
- Private endpoints give the resource a private IP inside your VNet (true private connectivity).
- Private endpoints are preferred for strict isolation and no public exposure.
- Choose private endpoints for compliance-driven designs.

Q278. Connections work from Azure but not from on-prem. What do you check?

Answer:

- Confirm hybrid connectivity (ExpressRoute/VPN) to the VNet hosting the private endpoint.
- Verify on-prem DNS resolves the SQL FQDN to the private IP (DNS forwarding to Azure Private DNS).
- Check NSGs/firewalls along the path.
- Validate the required ports are open end to end.

Q279. How do redirect vs proxy connection policies affect performance?

Answer:

- Redirect connects the client directly to the node after an initial handshake — lower latency.
- Proxy routes all traffic through a gateway — higher latency but simpler port requirements.
- Use redirect for in-Azure/VNet clients (open the required port range).
- Proxy may be needed when only the gateway port can be opened.

Q280. A Managed Instance's subnet is misconfigured and provisioning fails. What are the requirements?

Answer:

- MI needs a dedicated subnet delegated to Managed Instance with sufficient address space.
- The subnet must not host other resources and must allow required management traffic.
- Apply the correct NSG/route table per MI networking rules.
- Fix the subnet configuration before redeploying.

Q281. How do you connect multiple VNets/apps to the same database securely?

Answer:

- Use a private endpoint and VNet peering/hub-spoke so peered VNets resolve the private IP.
- Centralize DNS resolution via a hub Private DNS zone.
- Apply NSGs to control which subnets can reach the endpoint.
- Avoid re-enabling public access.

Q282. A firewall rule of 0.0.0.0 was found on a production server. Why is that bad and what do you do?

Answer:

- It effectively allows all Azure-internal traffic — far too permissive for production.
- Remove it and rely on private endpoints or specific IP/VNet rules.
- Enforce via Azure Policy to prevent recurrence.
- Audit access logs for any misuse while it was open.

Q283. How do you troubleshoot intermittent connectivity/latency to Azure SQL?

Answer:

- Check for transient PaaS reconfiguration errors (expected) and confirm app retry logic.
- Measure network latency/packet loss on the path (VPN/ExpressRoute).
- Verify DNS resolution and connection policy (redirect vs proxy).
- Inspect NSG/firewall and any proxy hairpinning.

Q284. What ports and endpoints matter for Azure SQL connectivity?

Answer:

- Gateway traffic uses 1433; redirect policy also needs the 11000-11999 range open.
- Private endpoints route over private IPs within the VNet.
- Ensure outbound rules from the client subnet permit these.
- For MI, follow the specific subnet/NSG port requirements.

Q285. How do you ensure DNS works correctly with private endpoints across the org?

Answer:

- Link the Private DNS zone (e.g., privatelink.database.windows.net) to all relevant VNets.
- For on-prem, forward the zone to Azure DNS via a resolver/conditional forwarder.
- Validate FQDN resolves to the private IP everywhere clients live.
- Avoid split-horizon mismatches that break resolution.

Q286. An app in one region must fail over to a database in another. Networking implications?

Answer:

- Ensure private endpoints/DNS exist in both regions for the failover-group members.
- The failover-group listener stays stable, but DNS must resolve to the active private endpoint.
- Configure cross-region VNet connectivity as needed.
- Test failover including the network path, not just the database role change.

Q287. How do you secure connectivity for a SQL VM's management plane?

Answer:

- Use Azure Bastion or Just-in-Time VM access instead of open RDP.
- Keep SQL on a private subnet with NSGs restricting the port.
- No public IP on the SQL VM unless strictly required.
- Monitor and alert on management access.

Q288. A partner needs limited access to one database over the internet temporarily. Safer approach?

Answer:

- Prefer a private link/VPN over public exposure if at all possible.
- If public is unavoidable, restrict to the partner's specific IP range and time-box it.
- Enforce TLS and Entra/strong auth.
- Audit access and remove the rule when done.

Q289. How do you validate the entire network path before a go-live?

Answer:

- Test connectivity from every client subnet/region to the endpoint.
- Confirm DNS resolution, port reachability, and latency meet expectations.
- Simulate failover and re-test the path.
- Document the topology and required rules.

Q290. Traffic must stay on the Microsoft backbone and never traverse the public internet. How do you guarantee it?

Answer:

- Use private endpoints for the databases and ExpressRoute/VPN for hybrid links.
- Disable public network access on the servers/MIs.
- Enforce via Azure Policy across the estate.
- Verify with connectivity tests and flow logs that no public path is used.

13. Cost Optimization, Licensing & Governance

DOMAIN 13

Cost Optimization, Licensing & Governance

10 real-time scenarios

Q291. Leadership asks how to cut the Azure SQL bill without hurting performance. What levers do you pull first?

Answer:

- Right-size tiers from real utilization instead of peak-of-peak provisioning.
- Apply Azure Hybrid Benefit to reuse existing SQL licenses.
- Buy reserved capacity (1 or 3 years) for steady workloads.
- Use Serverless/auto-pause for intermittent dev/test databases.

Q292. How does Azure Hybrid Benefit reduce cost and where does it apply?

Answer:

- It lets you apply existing SQL Server licenses with Software Assurance to Azure SQL MI, VM, and vCore databases.
- This removes the license portion of the compute cost (significant savings).
- It requires eligible licenses and correct license-type configuration.
- Track usage to stay compliant with license entitlements.

Q293. When do reserved capacity purchases make sense for databases?

Answer:

- For steady, always-on workloads where the vCore/tier is stable for 1-3 years.
- Reservations discount the compute significantly versus pay-as-you-go.
- They stack with Azure Hybrid Benefit for maximum savings.
- Avoid over-committing on workloads likely to change tier soon.

Q294. A monthly bill spiked unexpectedly. How do you find the cause?

Answer:

- Use Cost Management to filter by resource and break down by meter (compute, storage, backup, geo-replication).
- Look for tier scale-ups, added replicas, LTR growth, or Serverless not pausing.
- Correlate with recent changes/deployments.
- Recommend corrective action (right-size, reservations, fix auto-pause).

Q295. How do you govern spend so teams can't over-provision?

Answer:

- Use Azure Policy to restrict allowed tiers/regions and require tags.
- Set budgets and cost alerts per subscription/resource group.
- Enforce deployments through IaC/pipelines with review.
- Report cost-per-team via tags for accountability.

Q296. Backup storage cost is growing. How do you optimize it?

Answer:

- Right-size PITR retention and review LTR policies (long yearly retention is expensive).
- Choose zone/locally-redundant backup storage where residency allows instead of geo-redundant.
- Remove unnecessary long-term backups per governance.
- Monitor backup storage against the included allowance.

Q297. Dev/test environments run 24/7 unnecessarily. What's the cost fix?

Answer:

- Move to Serverless with auto-pause or automate scale/stop off-hours.
- Use the Dev/Test subscription offer/pricing where eligible.
- Right-size dev tiers well below production.
- Tag and automate lifecycle to avoid orphaned resources.

Q298. How do you decide between DTU and vCore purely on cost?

Answer:

- vCore unlocks Hybrid Benefit and reservations — usually cheaper for owned licenses and steady load.
- DTU can be marginally simpler/cheaper only for very small stable databases.
- Model both against the actual workload before deciding.
- For most production, vCore wins on cost flexibility.

Q299. A stakeholder wants a chargeback model per application. How do you enable it?

Answer:

- Enforce cost-allocation tags (app, cost-center, environment) via Policy.
- Use Cost Management grouped by tag to produce per-app spend.
- Automate monthly chargeback reports.
- Reconcile shared resources (pools) with a fair allocation method.

Q300. How do you keep licensing compliant when using Hybrid Benefit across many resources?

Answer:

- Track the number of cores/licenses applied vs owned entitlements.
- Use the SQL IaaS extension/license-type settings to manage AHB on VMs.
- Audit AHB usage periodically to avoid over-claiming.
- Document entitlements and mappings for true-ups/audits.

End of Document

Compiled by Praveen Madupu (PMSQLDBA) - sqldbachamps.com. 300 real-time Azure DBA scenarios across 13 domains.